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Efficient and scalable reinforcement learning for large-scale network control

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The primary challenge in the development of large-scale artificial intelligence (AI) systems lies in achieving scalable decision-making—extending the AI models while maintaining sufficient performance. Existing research indicates that distributed AI can improve scalability by decomposing complex tasks and distributing them across collaborative nodes. However, previous technologies suffered from compromised real-world applicability and scalability due to the massive requirement of communication and sampled data. Here we develop a model-based decentralized policy optimization framework, which can be efficiently deployed in multi-agent systems. By leveraging local observation through the agent-level topological decoupling of global dynamics, we prove that this decentralized mechanism achieves accurate estimations of global information. Importantly, we further introduce model learning to reinforce the optimal policy for monotonic improvement with a limited amount of sampled data. Empirical results on diverse scenarios show the superior scalability of our approach, particularly in real-world systems with hundreds of agents, thereby paving the way for scaling up AI systems.

Achieving scalable decision-making becomes a critical challenge when deploying artificial intelligence (AI) models into large-scale systems[1]. First, this requires effective information exchange among system entities to help an agent perceive the state of the environment and other agents[2]. However, owing to constraints and high costs associated with communication[3], achieving comprehensive information sharing across the entire system becomes unfeasible (Fig. 1a). For example, in traffic networks, frequent and extensive communication between traffic lights can cause a substantial power loss (Fig. 1b). In some systems involving user information, centralized information collection also increases the risk of privacy leakage[4]. Second, efficient sample acquisition and learning from limited data are necessary because the cost of agent–environment interaction rises exponentially with the scale of the system[5], and in some scenarios, only limited interactions are permitted. Therefore, developing an effective communication mode and a sample-efficient approach is crucial for achieving scalable decision-making.

As an advanced paradigm of distributed AI, multi-agent reinforcement learning (MARL) provides a possible solution[6,7] and has made progress in various scenarios, including autonomous driving[8–12], wireless communications[13,14], multi-player games[15,16], power systems[17,18] and urban transportation[19–24]. The advantage of MARL lies in its ability to perform nonlinear fitting solely by data and attain efficient inference. In contrast, traditional methods such as model predictive control (MPC) require precise system dynamics[25,26], which are often difficult to obtain in complex systems. Even with accurate dynamics, traditional methods often rely on system linearization[27], which inevitably harms performance by disregarding numerous intricate nonlinear factors and system perturbations inherent in complex systems. In addition, deficiencies in computational efficiency, numerical stability and communication costs

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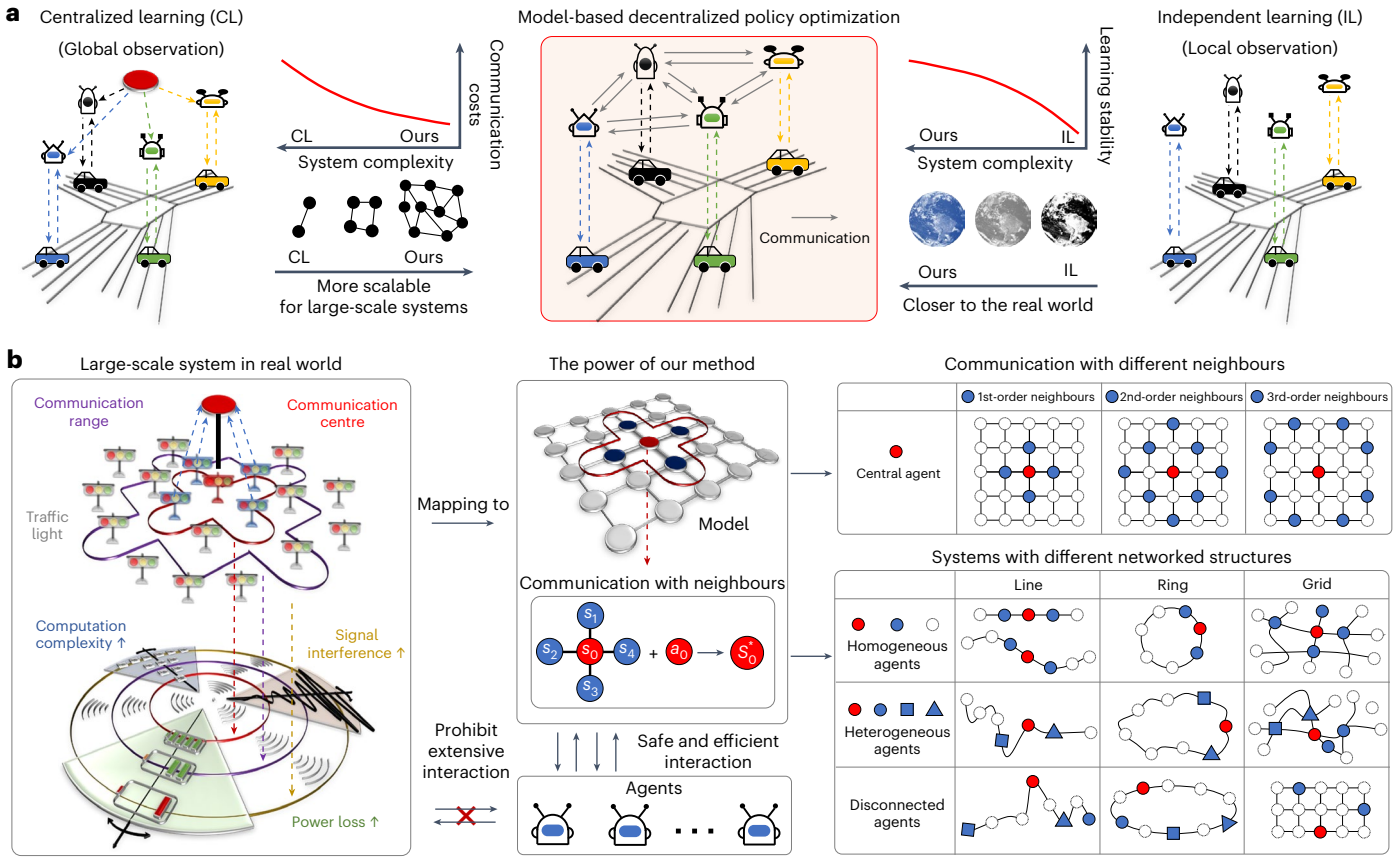


Fig. 1 | Research motivation and relationship of networked agents. a, When comparing our method with two main learning paradigms, centralized learning (left) requires each agent to have global observations, significantly increasing algorithm complexity and communication costs, and reducing scalability. In independent learning (right), each agent learns in a single-agent fashion through local observations, which reduces system and algorithm complexity. However, the learning process is unstable, leading to poor decision-making performance and mismatch with real-world scenarios. Our method (middle) learns based on local communication between topologically connected agents, avoiding the drawbacks of these two learning paradigms and achieving superior performance with low observation costs. **b**, When controlling traffic lights on a traffic system, obtaining information from many other traffic lights to make decisions for many other traffic lights can lead to power loss with each communication step, thereby reducing the endurance of the traffic lights. In addition, frequent and large-scale communication operations increase the probability of signal interference and raise computational complexity, which are drawbacks of extensive communication operations. Our approach relies on local communication between topologically adjacent agents to reduce communication costs, power consumption and computational complexity. Each agent receives the state observation from its neighbours and aggregates them with its state and action to obtain the final decision-dependent information. s_0 and a_0 represent the state and action, respectively, of the agent itself; s_1–s_4 represent the states of four adjacent agents and s_0* represents the aggregated information state of the agent. Furthermore, our framework maps the real system to a model, allowing it to learn policies through safe and efficient interactions between agents and the model. On the right, we describe the relationships between agents and neighbours of different orders in our methodology, where the order corresponds to κ, and a larger κ implies a broader communication range. In our method, each agent accurately estimates the global value and policy gradient solely through its neighbours' information, aiding policy learning. Another advantage of our solution is its ability to handle heterogeneous agents and systems of multiple types. In our experiments, CACC represents a linear-type system, Flow represents a ring-type system, and ATSC, Power Grid, Real Power-Net and Pandemic Networks represent grid-type systems. The agents in CACC, Flow and ATSC-Grid are homogeneous, while agents in ATSC-Monaco, ATSC-New York, Real Power-Net and Pandemic Networks are heterogeneous. In addition, in the non-adjacent setting, some disconnected agents existed in systems.

are also difficult to avoid for traditional methods[28]. Therefore, we focus on MARL approaches to realize large-scale AI systems.

However, despite its numerous advantages, developing an efficient and scalable MARL algorithm applicable to real-world scenarios remains challenging[29–36]. Previous works towards alleviating information exchange have been shown to be insufficient because they rely on impractical assumptions and redundant communication mechanisms, resulting in inferior adaptability to complex systems compared with our framework. Specifically, one direction involves connecting agents through a network topology and executing decentralized control by neighbouring information[21,37–39]. On the basis of this setting, existing methods have achieved policy learning convergence by assuming linear approximation of the value functions[40], or assuming that the system dynamics are fully independent and decouplable[39]. By contrast, we discard this assumption to adapt to more generalized systems. In addition, some insightful research adopted the truncated policy gradient method to optimize local policies and provided guarantees for policy improvement where communication is limited[41,42]. We further extend the concept of truncated Q-learning[41] (Q represents the value of state-action pairs) from the tabular case to deep MARL, thereby factoring global Markov decision processes (MDPs) into marginal transition to reduce system complexity[43]. This extension provides theoretical guarantees for policy improvement at each step in general systems. Another direction involves learning hidden information through computational communication protocols[21,44–47], whereas our approach requires only neighbour states. This indicates that each agent needs less information for learning and inference[21].

Furthermore, high sample efficiency is also crucial yet unaddressed, rendering current methods effective in only small-scale or simulated environments[16,48–50]. Model-based methods are widely studied as a promising approach for improving sample efficiency[51–56], as they can leverage estimated dynamic models to execute efficient interactions in for acquisition of policy learning data[52,57]. This is particularly helpful for the application of MARL in large-scale systems with constrained interactions (Fig. 1b). However, previous research on model-based MARL is primarily restricted to specific scenarios, for example, two-player zero-sum games[58,59], the pursuit-evasion game[60] or tabular tasks[61]. Some works formulate opponent models or dynamics in a decentralized structure[62], but these approaches assume full-state observability, and their scalability has not been sufficiently verified. Subsequently, some researchers extended model-based prioritized sweeping to a MARL scenario[61] but were limited to only the tabular MARL setting, and thus were unable to address more general tasks. In summary, it is difficult to achieve scalable decision-making with existing model-based MARL methods owing to the following limitations[63]. (1) Despite being rooted in local models, they still rely on a global critic, undermining their effectiveness in communication-constrained scenarios[64–66]. (2) Theoretical analysis and bounded estimation for multi-agent model learning are lacking, resulting in an insufficient understanding of the model's impact on MARL[65,67]. (3) The integration of model learning and policy learning in MARL scenarios, a potential iterative process in real-world scenarios, has not been achieved[65].

Here we surpass the existing research bottleneck and take a significant step towards a decision paradigm that is scalable to large-scale systems. First, we propose the ξ-dependent networked MDP, which has the Markovian property of transition. ξ characterizes the difference between the decoupled system dynamics and the real system dynamics. This is a form of decentralized partially observable MDP, and its advantage lies in its ability to deal with more general partially observable cooperative multi-agent tasks. This concept bridges the gap between standard network systems and general multi-agent systems, providing the necessary theoretical framework and analytical tools for research into multi-agent systems. Second, building on ξ-dependent network systems, we propose a decentralized model-based framework for networked systems, called model-based decentralized policy optimization. Our method is efficient and scalable, enabling overall performance improvement of the general system even when individual agents have severely limited access to information retrieval (our method can achieve the minimum local message size $O(|s_i|)$). O is a mathematical notation used to describe the asymptotic upper bound of the complexity of an algorithm; s_i rerepresents the state of agent i. In our method, we couple localized model learning with decentralized policy optimization. Specifically, we use well-learned localized models to predict future states and use local communication to broadcast their predictions. To alleviate the issue of compounding model error, we adopt a branching strategy[54,68] by replacing few long-horizon rollouts with many short-horizon rollouts to reduce compounding errors in model-generated rollouts. In the policy optimization part, we perform decentralized proximal policy optimization (PPO)[69] using data generated by localized models and incorporating an extended value function. Theoretically, we prove that the policy gradient computed from the extended value function is a close approximation to the true gradient, establishing the theoretical bridge between model learning and monotonic policy improvement. Our theoretical analysis begins with extending the fully independent systems[41,70] to more realistic and general multi-agent systems, and expanding single-agent model learning theory[54] to multi-agent systems.

Empirically, we evaluate our algorithms in highly realistic simulators and real-world scenarios in a progressive manner, encompassing various systems such as transportation, electric energy and social healthcare. Specifically, the evaluation scenarios involved: (1) cooperative adaptive cruise control (CACC)[21] and connected autonomous vehicle control (Flow)[71], (2) adaptive traffic signal control (ATSC), including the construction of a traffic network using real maps of Monaco[21] and New York, (3) IEEE Power Grid[72] and the real power systems constructed using real-time electricity consumption data from Portugal[73], and (4) the Pandemic Networks constructed by epidemiologists based on real epidemiological dynamics, COVID-19 prevention and control policies, and infection status from real Swedish government data[74]. These tasks accurately and comprehensively simulate the complexities and challenges that exist in large-scale networked systems, with agent counts reaching 199 and 436. Importantly, although some MARL methods can handle tasks involving large-scale multi-agent systems (500 agents and above)[75–77], these approaches often simplify the complexity of large-scale problems through agent grouping[76,77] or mean-field approximations[75]. The problem setting remains approximately centralized and cannot address decentralized multi-agent learning in communication-limited scenarios. In contrast, our method is decentralized. Therefore, it is necessary to qualify our statements to indicate that our method significantly surpasses previous decentralized MARL methods in terms of the number of agents it can handle.

Model-based decentralized policy optimization

In the general framework of our algorithm (Algorithm 1), three key components need to be specified: (1) how to perform decentralized policy optimization, (2) how to query localized models and (3) to what extent can we approximate the true policy gradients? To train the models and policies, we maintain two experience buffers, $\mathcal{D}^{\mathrm{E}}$ for trajectories generated by true environment p and $\mathcal{D}^{\mathrm{M}}$ for data generated by the model. The trajectories for agent i consist of $(s_i, a_i, s_i', r_i, d_i)$, $i \in \mathcal{V}$, where $\mathcal{V} = \{1, \ldots, n\}$ is the set of agents, s_i is state, a_i is action, s_i' is next state, r_i is reward and d_i is binary, indicating whether the task is completed or reaches maximum episode length. The architecture of our framework is presented in Fig. 2a. Below we present the details of model learning and policy improvement.

For each agent i, let π_i^* denote the optimal policy and $V_i(s)$ denote the critic. The parameterized policy π^{θ_i} with parameters θ and critic V^{ϕ_i} with parameters ϕ are used to approximate these functions. Let $\{(s_{i,\tau}, a_{i,\tau}, r_{i,\tau})\}_{i\in\mathcal{V},\tau\in\mathcal{B}}$ represent a mini-batch of samples from $\mathcal{D}^{\mathrm{M}}$ generated by policies $\pi^{\theta_i}, i \in \mathcal{V}$, where τ is the collected interaction trajectory and $\mathcal{B}$ represents a mini-batch sampled from the experience buffer. As it is not easy to obtain the true value $V_i(s)$, we adopt the extended value function[39], which is defined as

$$V_i(s_{N_i^\kappa}) = \mathbb{E}_{s_{N_{-i}^\kappa}}\left[\sum_{t=0}^{\infty} r_i^t | s_{N_i^\kappa}^0 = s_{N_i^\kappa}\right], i \in \mathcal{V}. \tag{1}$$

Let N_i^κ denote the κ-hop neighbour of agent i. Note that $V_i(s_{N_i^\kappa})$ is a good approximation of $V_i(s)$, with the discrepancy decreasing exponentially with κ (according to Theorem 3). To generate the objective for the extended value function, or return R_i, we use reward-to-go technique[78]. The target of $V_i(s_{N_i^\kappa}^t)$ is

$$R_i^t = \sum_{l=0}^{T-t-1} \gamma^l r_i^{t+l} + V^{\phi_i}\left(s_{N_i^\kappa}^T\right), \tag{2}$$

where $\gamma \in (0, 1)$ is the temporal discount factor and T is the length of each episode. The loss for learning value function approximation is defined as

$$\mathcal{L}(\phi_i) = \frac{1}{|\mathcal{B}|}\sum_{\tau\in\mathcal{B}}\left(V^{\phi_i}\left(s_{N_i^\kappa}^\tau\right) - R_i^\tau\right)^2. \tag{3}$$

Empirically, we use communication within κ-hop neighbours to generate an estimation of the global value function, denoted as

$$\tilde{V}_i\left(s_{N_i^\kappa}^t\right) = \frac{1}{n}\sum_{j\in N_i^\kappa} V_j\left(s_{N_j^\kappa}^t\right) \tag{4}$$

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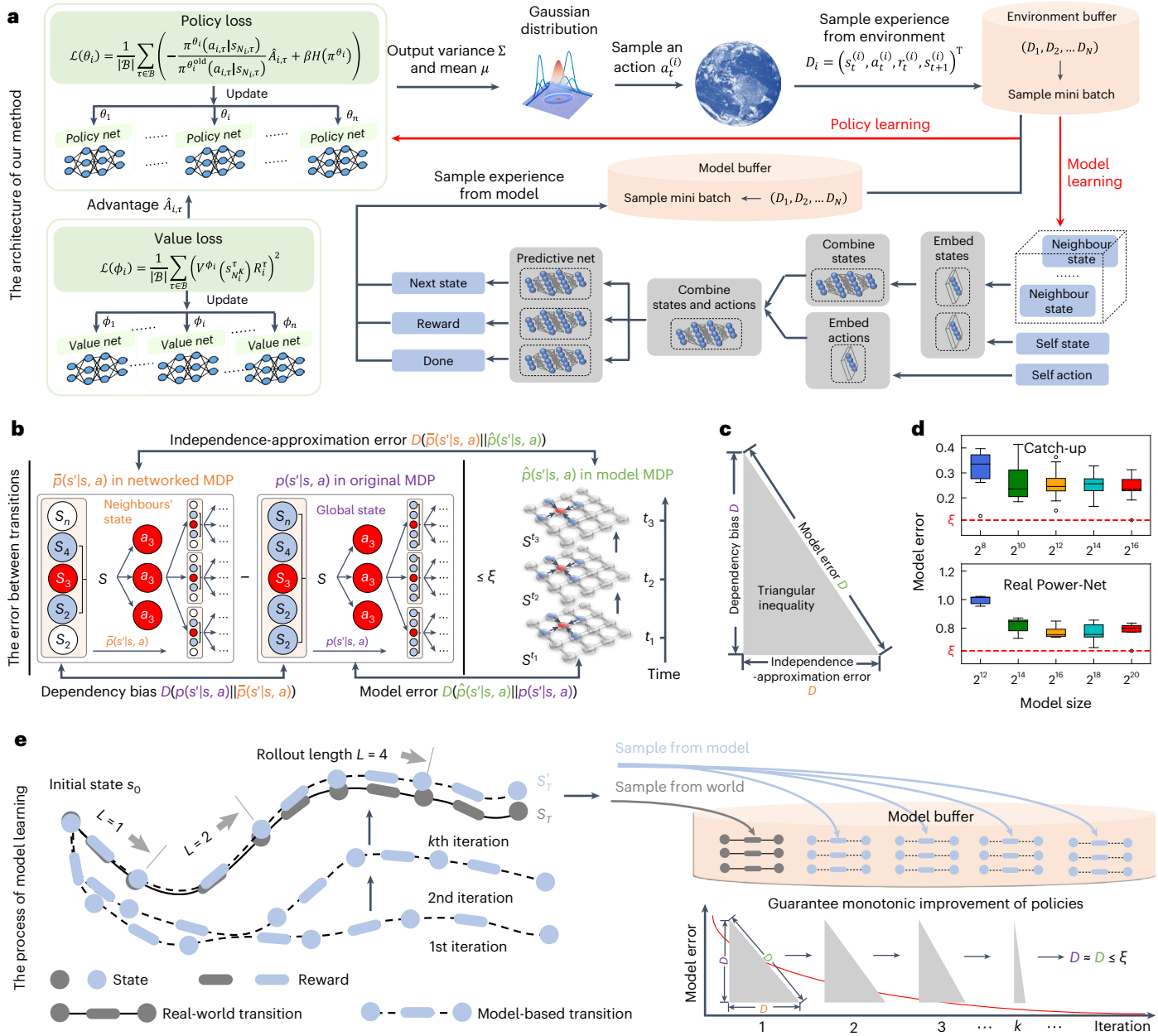


Fig. 2 | Architecture of our method, distinctions and connections among different MDPs, and the model learning process. a, Our algorithm comprises decentralized policy, value function, model, model buffer and environment buffer. The red arrows highlight primary steps, involving experience sampling from both the model and the real environment for policy and model learning. The model framework incorporates embedding layers for state and action feature extraction, followed by merging based on graph network topology. **b**, Distinction and connection of different p in networked MDP, original MDP and model MDP as depicted in inequality (12). The differences between different p constitute dependency bias, model error and independence-approximation error, respectively. **c**, These different D constitute the triangular inequality relationship in inequality (12). The main objective of model learning is to minimize the independence-approximation error $D(\bar{p} \parallel \hat{p})$ to reduce the difference between model error $D(p \parallel \hat{p})$ and dependency bias $D(p \parallel \bar{p})$. **d**, In real-world complex systems, we prove that the values of ξ are all small positive values ($\xi \geq 0.2$ in Catch-up and $\xi \geq 0.8$ in Real Power-Net), which proves the authenticity and correctness of our assumptions about ξ-dependent systems in Definition 2. The data ($n = 5$) are presented as median values (the central line of each box) along with the 25th and 75th percentiles (the bottom and top edges of each box), minima and maxima (the whiskers attached to each box), and outliers (outside the box and whiskers). **e**, The process of model learning in our approach involves sampling trajectories from the model buffer for multiple iterations of learning (S_T represents the real state when the system has reached step T, and S_T' represents the predicted state of the system when it reaches step T by the model). As a result, the model's predictions of transition gradually approach the real world, with $\sup D(p \parallel \hat{p}) \approx \sup D(p \parallel \bar{p}) = \xi$, which promotes a monotonic improvement of the policies.

where t represents a certain time step during the networked MDP, and j represents the agent index adjacent to agent i in κ-hop adjacency. The advantage is thus defined as $\hat{A}_t(s^t_{N_i^\kappa}, a^t_i) = \mathbb{E}_{s^{t+1}_{N_i^\kappa}}\left[r^t_i(s^t_{N_i^\kappa}, a^t_i) + \gamma \tilde{V}_i(s^{t+1}_{N_i^\kappa}) - \tilde{V}_i(s^t_{N_i^\kappa})\right]$. We adopt a PPO agent[69] in implementations. The loss function of an agent in our method is defined as:

$$\mathcal{L}(\theta_i) = \frac{1}{|\mathcal{B}|}\sum_{\tau \in \mathcal{B}}\left(-\frac{\pi^{\theta_i}(a_{i,\tau}|s_{N_i,\tau})}{\pi^{\theta_i^{\text{old}}}(a_{i,\tau}|s_{N_i,\tau})}\hat{A}_{i,\tau} + \beta H(\pi^{\theta_i})\right). \quad (5)$$

where $H(\pi^{\theta_i})$ represents the policy entropy function used to compute the entropy of the policy π^{θ_i}, and β is the constant coefficient corresponding to the policy entropy.

In general, larger κ leads to higher performance on policy training, but also more communication costs. In algorithms, computation cost cannot be overlooked, as larger κ results in more complex models or policy architectures, thereby complicating the training process. We discuss more about this problem in the experiments section.

Update model

To perform decentralized model-based learning, each agent maintains a localized model. This localized model observes the state of its κ-hop neighbours and its own action, with the goal of predicting the next time step's information, including state and reward. Let $\hat{s}_{i,t+1} = p^{\psi_i}(s_{N_i,t}, a_i)$; here we parameterized the neural network model p, with parameters represented by ψ. In practice, each agent stores the data locally. The models are updated by minimizing the following objective function:

$$\mathcal{L}(\psi_i) = \frac{1}{|\mathcal{B}|} \sum_{\tau \in \mathcal{B}} \| \hat{s}_{i,\tau+1} - s_{i,\tau+1} \|^2. \quad (6)$$

Scaling model-based methods to real tasks can lead to decreased performance, even if the model itself is relatively accurate. A primary reason for this is the compounding modelling error during long model rollouts, where errors accumulate along the rollout trajectory, ultimately making it inaccurate. To mitigate the negative effects of model error, we adopt a branched rollout scheme[54,68]. In branched rollout, the model rollout does not start from an initial state, but from a state randomly selected from the most recent environmental trajectory τ. In addition, the model rollout length is fixed to T. The model training steps are described in lines 4–7 of Algorithm 2.

Results

In this section, we introduce the results of our method and baselines (more details in Supplementary Table 1) in the control of large-scale networked systems (more description in Supplementary Section 2). We have solved many control problems in simulators and real-world systems (more details in Supplementary Table 2). Baselines consist of two main categories: model-based algorithms such as MAG[64], independent model-based policy optimization (IMPO) and MPO_{na}, and model-free algorithms such as CommNet[47], IC3Net[79], DIAL[45], NeurComm[21], ConseNet[40], FPrint[80], CPPO and DPPO, where IMPO is a fully independent model-based MARL method, MPO_{na} is an ablated disconnected agents in our method (Fig. 1b) and our method with linear models ablated linear models in our method (more details in Supplementary Fig. 1). Overall, our method has decentralized models, policies and value functions, with the minimum local message size. Especially compared with the centralized methods, the communication costs of our method in different scenarios is only about 5–35% of that in centralized methods (more details in Supplementary Table 3).

Training results and ablation study

Learning process. We first focus on the convergence of the episode reward and sample efficiency in the learning process. Figure 3e shows that our method significantly improves sample efficiency and achieves the highest sample efficiency, converging within 1×10^{-5} training steps, exceedingly outperforms baselines (more results in Supplementary Fig. 2). The reason is that the usage of the model will greatly increase the amount of experience to support training and improve the exploration efficiency (corresponding to lines 9–12 in Algorithm 3). As seen in Fig. 3d, our algorithm generates more extensive state-visited (the state-visited point set covers more areas in two-dimensional space), further confirming this point. Therefore, the experimental results are consistent with the theoretical results. From the results in different tasks, we observe that the performance of CPPO does not exceed that of the algorithms that use extended value functions. In this way, we conclude that by using an extended value function, a centralized algorithm can be decomposed into a decentralized algorithm, but the performance would not drop significantly. According to Theorems 3 and 4, the extended value function is accurate enough to evaluate the overall multi-agent system and support the training process. Therefore, the experimental results are consistent with the theoretical results. Compared with the two model-based baselines IMPO and MPO_{na}, we conclude that the existence of the extended value increased the estimation accuracy of the global value function and gradient. Although theoretical analysis of our method in Theorems 3 and 4 does not provide more restrictions on whether the neighbour selection is adjacent, in the actual experiment we recommend using information of adjacent neighbours in the topology of the networked systems to predict the global value and gradient, which can reduce the variance of the learning process and improve the performance. It is worth noting that MAG models the learning process of model errors as MARL tasks, resulting in lower model errors in certain scenarios. However, MAG still relies on global modelling information during the learning phase. In the model prediction phase, the traditional control algorithm MPC is utilized in the model prediction phase, thus having a higher computational complexity and communication cost.

In terms of optimal performance, Fig. 3a shows that our method reaches the highest performance with the minimum local message size among all the baselines. Due to the fact that CPPO is a fully centralized algorithm, where each agent obtains global complete information, it has the highest theoretical performance and performs well on ATSC-Monaco. However, due to its dependence on large network size and computing resources, it is difficult to truly achieve theoretical optimal performance. MAG also has high performance in multiple scenarios but leads to high computational burden and poor scalability.

Rollout length. To verify the relationship of rollout length k to Theorems 1 and 2, we designed relevant experiments to observe the effects of different choices of rollout length on the convergence of the episode reward and sample efficiency. Model rollout length represents the step size of using the model forward to predict the state and reward. Figure 3b shows more details of the experimental results. We choose four representative environments: CACC Catch-up, Ring Attenuation, ATSC-Monaco and Power Grid with the set of rollout lengths 1, 25, 50, 100 and 200, to validate how sensitive the accuracy of environment model is to rollout length. From the final episode reward and sample efficiency in Ring Attenuation under the rollout lengths of 1, 25 and 100, it can be concluded that the environment model is accurate enough to support normal training. However, the accuracy of model drops significantly when the rollout length is 200, which is reflected in the final episode reward and sample efficiency being lower than for shorter rollout lengths. According to Theorems 1 and 2, a shorter rollout length k leads to a lower model error and better convergence of episode reward. Therefore, the experimental results are consistent with the theoretical results.

κ-hop neighbours. To further verify the relationship of κ with Theorems 3 and 4, we designed relevant experiments to observe the effect of different choices of κ on the convergence of the episode reward, sample efficiency and the model error, which is defined as the mean squared error loss under the fixed network structures in our framework. From Fig. 3c, we conclude that neighbour information is accurate enough for a model to predict the next state in these environments, but the effect of different κ on state prediction is different. When the network structure is fixed, the sampling efficiency will increase and the error of our model in predicting the state will increase with increasing κ. But when κ is greater than a certain value, the sampling efficiency will decrease and the error of our model in predicting the state will increase due to the limited fitting ability of the network. The optimal choice of κ is 4 under our network structure in Ring Attenuation and 3 in CACC Catch-up. There is almost no difference between different choice of κ in Ring Attenuation, which indicates that the information of neighbours

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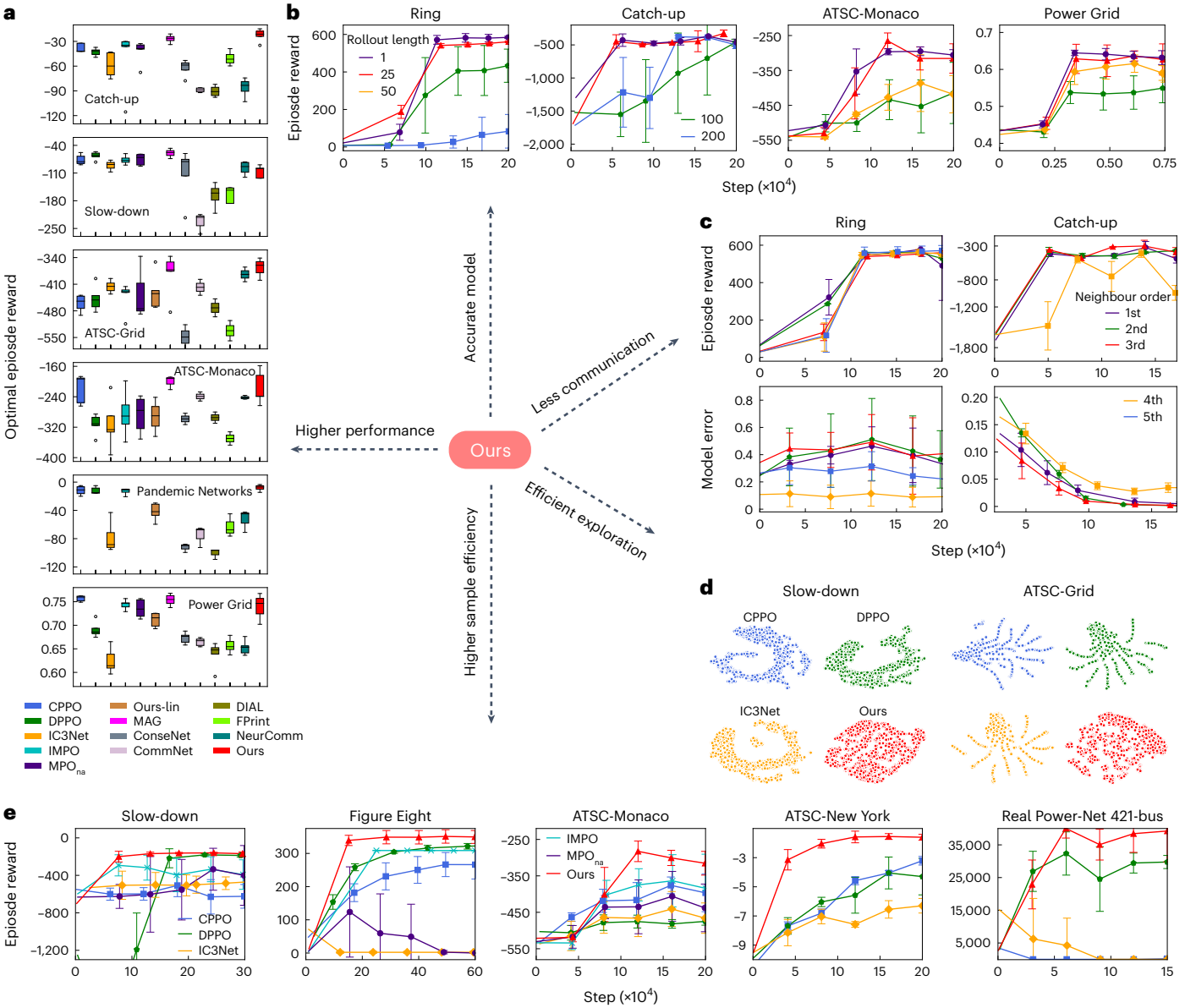


Fig. 3 | Main training results and ablation studies. These results demonstrate the advantages of our method, including high performance, accurate model learning, low communication cost, efficient exploration ability and higher sample efficiency. **a**, The performance upper bound of all the baselines in different scenarios with different random seeds. The red colour illustrates the performance of our algorithm. We compared it with model-based and model-free baselines. Our approach achieves superior performance in terms of optimality, as demonstrated by the upper edge of the red boxplot being at its highest position in the figure. In addition, our method shows excellent performance in the upper quartile, median and lower quartile, as shown in the plot. Furthermore, our algorithm demonstrates enhanced training stability, indicating the absence of outliers within the red boxplot. The data ($n = 5$) are presented as median values (the central line of each box) along with the 25th and 75th percentiles (the bottom and top edges of each box), minima and maxima (the whiskers attached to each box), and outliers (outside the box and whiskers). Ours-lin, our method with linear models. **b**, Training reward under different rollout lengths in Ring, Catch-up, ATSC-Grid and Real Power-Net 141-bus ($n = 5$). **c**, Training rewards and model error ($n = 5$) under different κ in Ring and Catch-up. Different κ correspond to different neighbour order (Fig. 1b). **d**, The states visited by different algorithms in Slow-down and ATSC-Grid. The distance between points represents the difference between states[84]. Due to the existence of the model, our approach outperforms other algorithms in exploration efficiency, thereby increasing sample efficiency. This augmentation is visually represented by the more even and expansive spatial coverage of the red data points within the depicted space. **e**, Training reward increment during the training process under different scenarios and different algorithms. Solid lines represent the mean rewards and error bars are s.d. ($n = 5$).

under $\kappa = 1$ is sufficient for policy learning. According to Theorems 3 and 4, a larger κ leads to a lower model error, higher sample efficiency and better convergence of episode reward. Therefore, the experimental results are consistent with the theoretical results.

Execution results

Connected vehicle control. Figure 4a,b shows the execution performance of the trained policies in Flow and CACC. In CACC, both the velocity and headway of vehicles with the same interval in the queue can be stably controlled around the target value by our method. For Flow, we conclude that our decentralized controller improves the efficiency of the overall traffic flow while ensuring safety (more results in Supplementary Fig. 3).

Traffic signal control. Figure 5a,b shows the scenario descriptions and the execution performance for ATSC-Grid, -Monaco and -New York. The

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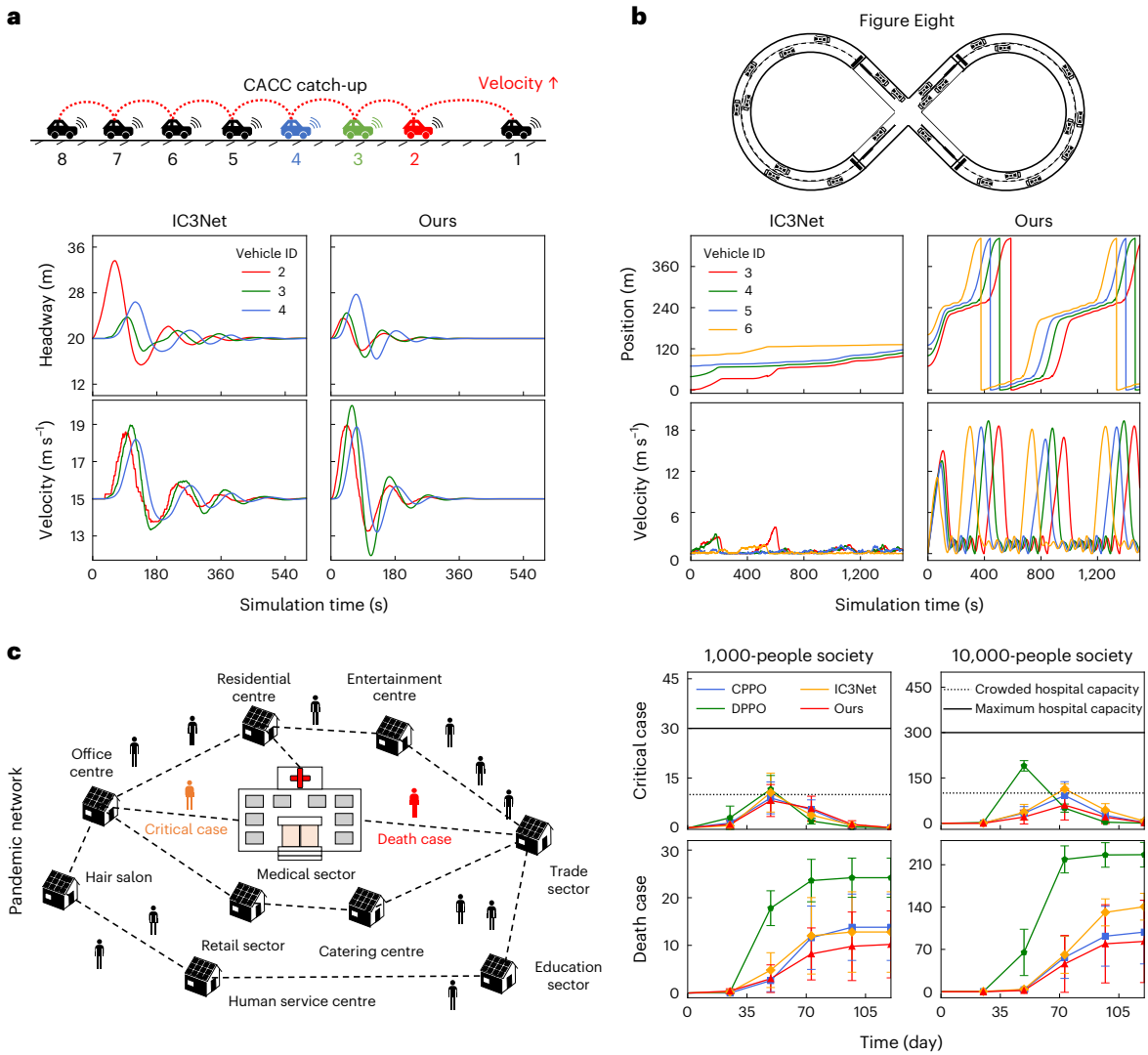


Fig. 4 | Evaluation results on vehicle control on CACC, Flow and pandemic network. a, Description of velocity and headway of vehicles 1, 5 and 8 in Catch-up. The headway stabilizes at 20 m, which matches the target headway. The velocity stabilizes at 15 m s^{-1}, matching the target velocity. **b**, Description of velocity and position of vehicles 1–4 in Figure Eight. The visualization of tasks is adapted from existing work[71]. Our approach controls the formation of queues to cross the intersection with an orderly process of accelerating to the target velocity and then decelerating to the safe velocity near 0 m s^{-1}. **c**, Description of a pandemic network, where various societal units adjust control policies to manage the critical cases and death cases. The right side shows the strategies provided by different algorithms for handling critical cases and death cases with different population sizes. The data are presented as mean ± s.d. The centre line represents the mean values and the error bar represents the s.d. ($n = 5$). Panel b adapted from ref. 71 under a Creative Commons licence CC BY 4.0.

average queue length and average intersection delay at intersections are important indicators to measure the level of traffic congestion. Compared with the baselines, our method significantly reduces the average queue length and intersection delay. Our solution still effectively solves traffic congestion with low communication costs, achieving the maximum number of arrived vehicles, average velocity and the minimum number of halting vehicles. The reason why our method in the New York City road network with 436 controllable intersections has a lower average velocity after 1,800 s is that in the later stage, our method has already cleared most vehicles to the destination, and only a few vehicles are driving at this time.

Pandemic network control. In the large-scale spread of an epidemic in society, real-time local decision-making is very important. As the infection situation gradually worsens, the decisions of different decision-making institutions (education departments, medical systems, social service systems and so on) usually rely on the local information possessed by the current institutions (Fig. 4c). We observe the overall infection of different infectious disease prevention and control strategies in a society with a growing population in three months. Figure 4c shows the relationship between critical cases and the maximum capacity of hospitals under different policies. If the number of critical cases exceeds the maximum capacity of the hospital, it indicates that the current virus transmission is very serious. Our approach can effectively control severe cases below the maximum capacity of the hospital and alleviate the pressure on medical staff. Figure 4c shows the relationship between death cases and maximum hospital capacity under different policies. Our method maintained a small number of deaths in multiple scenarios, controlling the mortality rate at a lower level between 0.8% and 1.5%, while also balancing regional economic development as much as possible (according to the final performance shown in Fig. 3a; more results in Supplementary Fig. 4).

Power control. Figure 6a shows the architecture of secondary voltage control in Power Grid. Figure 6c shows the execution performance of the trained policies based on different algorithms. We conclude that

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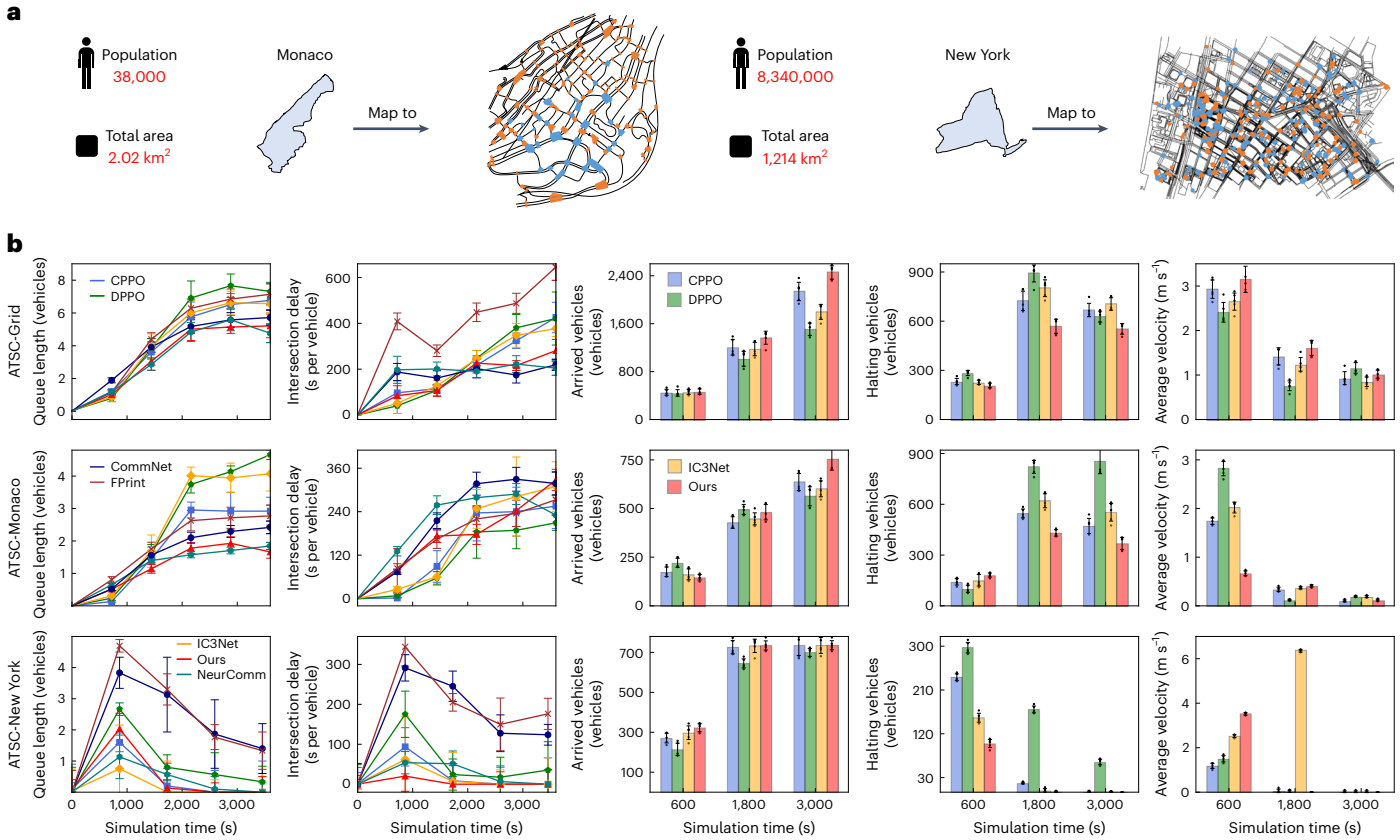


Fig. 5 | Evaluation results on large-scale traffic control from ATSC. a, Monaco traffic network (a part of Monaco, adapted from existing research[21]) with 28 heterogeneous traffic lights and New York traffic network with 432 traffic lights (a part of New York). **b**, Key evaluation indicators for ATSC tasks. The data are presented as mean ± s.d. The centre represents the mean values and the error bar represents the s.d. ($n = 5$). During the first 3,000 s, the system will continue to load vehicles, causing the traffic pressure at the intersections to gradually increase, and the average queue length will gradually increase. However, our method clears the traffic congestion well, so that the maximum average queue length does not exceed about 6 vehicles in Grid, 2 vehicles in Monaco and 1 vehicle in New York. This indicator is lower than other algorithms. After 3,000 s of vehicles loading, our method can maintain an average intersection delay at about 250 s per vehicle in Grid, 300 s per vehicle in Monaco and 10 s per vehicle in New York. Our method prevents it from increasing. At different times, our decentralized policies achieved the most arrived vehicles, the least halting vehicles and stable average velocity. It should be noted that in New York scenario, the actual total evaluation time is 1,000 s; to unify the time with all scenarios, we have standardized it to within 3,600 s. Credit: Monaco and New York maps in **a** created with GADM.org.

our method can maintain reliable control performance and adaptability to random disturbances while reducing the cost of communication (more results in Supplementary Fig. 5). Furthermore, we show the main results in the large-scale Real Power-Net 141-bus, 322-bus and 421-bus scenarios within 12 hours in Fig. 6b. Our method can maintain a higher safety control rate in the 141-bus scenario and achieve lower power loss in terms of q loss (the symbol q represents reactive power) and total power loss (Fig. 6d). When we expand the system scale to 322-bus (22 agents) and 421-bus (199 agents), the system complexity and modelling difficulty are higher. Our method still maintains effective policies, which demonstrate its scalability. The strong baseline OPF[81] algorithm, which relies on the real dynamics of the system, and its decision-making ability will decrease in large-scale systems. Besides, solving the constrained optimization problem in OPF is time-consuming, so it is difficult to react to the rapid change of load profile in a large network. This further confirms the advantages of our solution in practical power grid problems and modelling complex non-stationary dynamic systems.

Discussion

Here we provide a theoretical framework for scalable decision-making and propose a MARL framework for the control of general systems. Notably, our MARL framework remains effective even when the system size scales up to the hundreds of agents, which significantly surpasses the scale of problems addressed by previous decentralized methods. Therefore, this method holds promise in establishing large-scale AI decision-makers for traffic, energy systems and pandemic systems, thereby elevating the intelligence level in the real world. The core idea in this work shares an intriguing parallel with the theory of six degrees of separation[82]. Through interconnected topology, agents require only minimal information exchange to assess the global situation. In the future, we hope to explore the optimal system topology through information entropy theory and expand modules such as vision and natural language, which will comprehensively improve the ability of AI models and provide insights for developing scalable AI methodologies.

Methods

In this section, we mainly describe the details of the problem formulation and our algorithm[83] (more details in Supplementary Code 1), as well as some basic theoretical conclusions. First, we introduce networked MDP. We define an independent system[39] and ξ-dependent system, and we present diverse experimental results on algorithms and models.

Networked MDP

We consider a system with n agents as a graph. Specifically, n agents coexist in an undirected and stationary graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each agent in a system is represented as a node in an undirected and stationary graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, therefore $\mathcal{V} = \{1, \ldots, n\}$is the set of agents. $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$comprises the edges that represent the connectivity of all the agents. Each agent

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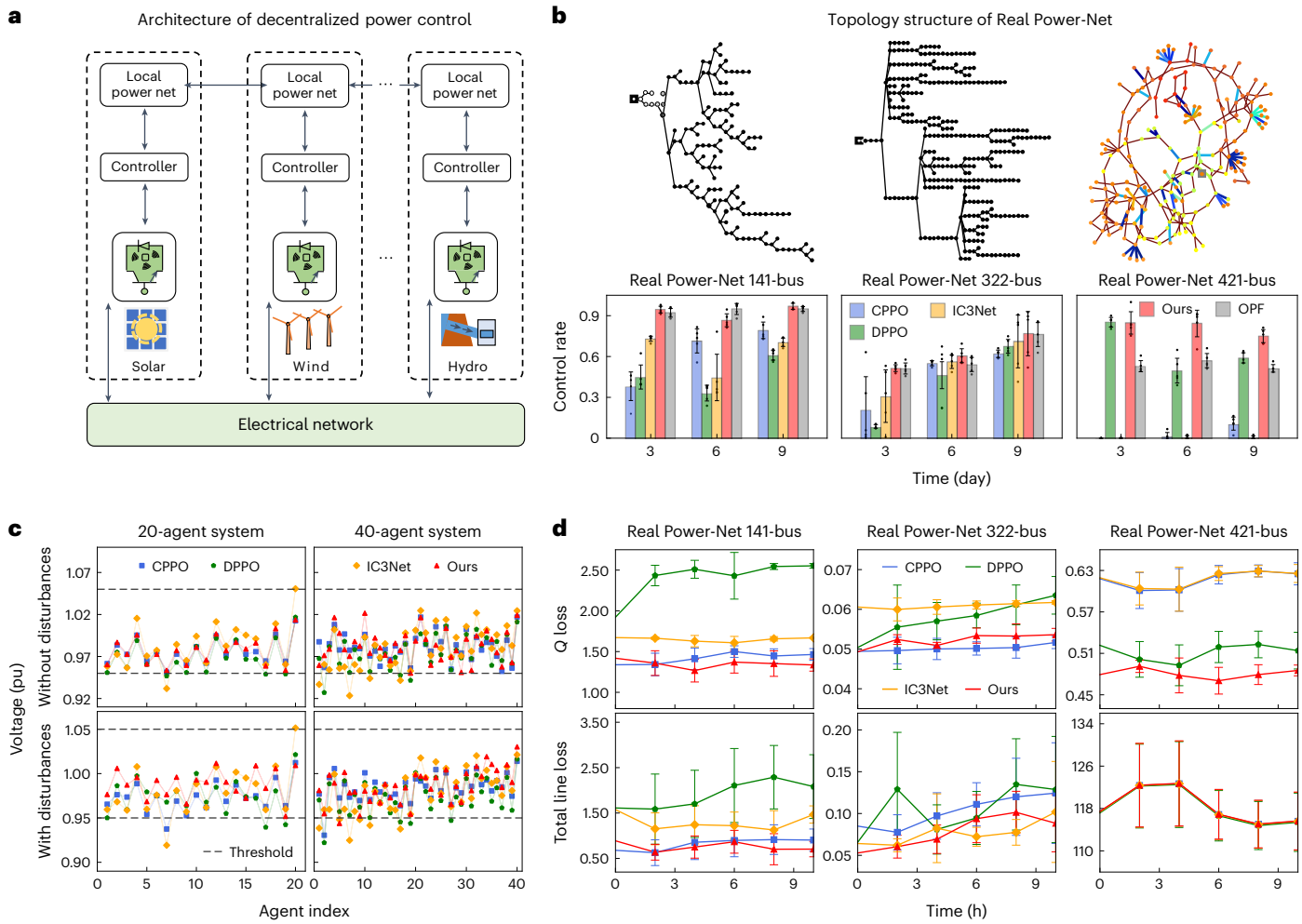


Fig. 6 | Evaluation results on large-scale power control from Power Grid and Real Power-Net. a, The architecture of secondary voltage control in Power Grid with the objective of stabilizing the output voltages of the distributed generators at a pre-defined reference value. 0.95 pu to 1.05 pu is the range of the reference value; pu (per unit) is a non-unit ratio used to represent the ratio of a voltage value to a reference value. This system structure is adapted from existing research[72]. **b**, The structures of 141-bus, 322-bus and 421-bus in Real Power0Net. We report the capabilities in terms of safety control rate. Control rate calculates the ratio of time steps. The data are presented as mean ± s.d. The top of the bar chart represents the mean values and the error bar represents the s.d. ($n = 5$) where all buses' voltages are under control for 12 hours. **c**, The voltage control performance without random disturbances and with random disturbances in Power Grid with 20 agents and 40 agents. The control performance of the baselines decreases when random disturbances are added and it is difficult to control the voltage of all distributed generators within the range of the reference value. But our method can control the voltage of most distributed generators well within the range of the reference value regardless of the presence of random disturbances. **d**, We report the Q loss and total line loss in Real Power-Net. The data are presented as mean ± s.d. The centre line represents the mean values and the error bar represents the s.d. ($n = 5$). Q loss represents the average of the mean reactive power generations by agents each time step during 12 hours. Total line loss represents the average of the total power loss overall buses per time step during 12 hours.

can communicate along the edges with their adjacent agents. Let N_i denote the neighbour of the agent i including itself. Let N_i^κ denote the κ-hop neighbour of i, that is, the nodes whose graph distance to i is less than or equal to κ. For the simplicity of notation, we also define $N_{-i}^\kappa = \mathcal{V} \setminus N_i^\kappa$. We define an adjacency matrix A with size of $n \times n$. If there is an adjacency relationship between agent i and agent j, then $A_{ij} = A_{ji} = 1$, otherwise $A_{ij} = A_{ji} = 0$. This adjacency relationship is determined by the system structure and κ.

We define its corresponding networked MDP as $(\mathcal{G}, \{\mathcal{S}_i, \mathcal{A}_i\}_{i\in\mathcal{V}}, p, r)$. Each agent i has its local state $s_i \in \mathcal{S}_i$, and performs action $a_i \in \mathcal{A}_i$. The global state is the concatenation of all local states: $s = (s_1, \ldots, s_n) \in \mathcal{S} := \mathcal{S}_1 \times \ldots \times \mathcal{S}_n$. The global action is $a = (a_1, \ldots, a_n) \in \mathcal{A} := \mathcal{A}_1 \times \ldots \times \mathcal{A}_n$. For the simplicity of notation, we define s_{N_i} to be the states of i's neighbours. The transition function is defined as: $p(s'|s, a) : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and the state transition satisfies Markov properties. Each agent maintains a localized policy $\pi_i^{\theta_i}(a_i|s_{N_i})$ with parameter $\theta_i \in \Theta_i$, which means that the localized policy is dependent on only states of its neighbours and itself. We use $\theta = (\theta_1, \ldots, \theta_n)$ to denote the tuple of localized policy parameters, and $\pi^\theta(a|s) = \prod_{i=1}^n \pi_i^{\theta_i}(a_i|s_{N_i})$ to denote the joint policy. The reward function for each agent depends on only local state and action: $r_i(s_i, a_i)$, and the global reward function is defined to be the average reward $r(s, a) = \frac{1}{n}\sum_{i=1}^n r_i(s_i, a_i)$. The goal of reinforcement learning is to find a policy π^θ that maximizes the accumulated reward:

$$\pi^{\theta^*} = \underset{\pi^\theta}{\operatorname{argmax}}\, \mathbb{E}_{\pi^\theta}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right], \tag{7}$$

where $\gamma \in (0, 1)$ is the temporal discount factor. The value function is defined as

$$V_{\pi^\theta}(s) = \mathbb{E}_{\pi^\theta}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)|s^0 = s\right] = \frac{1}{n}\sum_{i=1}^{n} V_i(s). \tag{8}$$

In the last step, we have defined $V_i(s)$, which is the value function for individual reward r_i.

Model-based learning. Let $\eta[\pi]$ denote the return of the policy in the true environment:

$$\eta[\pi] = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \tag{9}$$

Let $\hat{\eta}[\pi]$ denote the returns of the policy under the approximated model. To analyse the difference between $\eta[\pi]$ and $\hat{\eta}[\pi]$, we need to construct a bound

$$\eta[\pi] \geq \hat{\eta}[\pi] - C(\, p, \hat{p}, \pi, \pi_{\mathrm{D}}), \tag{10}$$

where C is a non-negative function, and π_{D} is the data-collecting policy. According to equation (10), if every policy update ensures an improvement of $\hat{\eta}[\pi]$ by at least C, $\eta[\pi]$ will improve monotonically. This inequality was first presented in single-agent domain[54].

ξ-dependent networked system

A networked system may have some extent of locality, meaning that in some cases local states and actions do not affect the states of distant agents. In such systems, environmental transitions can be factorized, and agents are able to maintain local models to predict future local states. We define an independent networked system (INS) and an ξ-dependent networked system as follows. Figure 1b shows different topological structures in networked systems. In addition, exponential decay has been shown to hold in networked MARL when the network is static[41,70]. This means that with the increase of the topological distance, the influence between agents will gradually decrease in networked systems. We have proved the properties of these networked systems through experiments in the Ring Attenuation (more results in Supplementary Fig. 6).

Definition 1. *An environment is an INS if:*

$$p(s'|s,a) = \prod_{i=1}^{n} p_i(s_i'|s_{N_i^\kappa}, a_i). \tag{11}$$

INS might be an assumption that is too strong to hold. However, for the dynamics that cannot be factorized, we can still use an INS to approximate it. Let D_{TV} denote the total variation distance between distributions, we have the following definition:

Definition 2. *(ξ-dependent) Assume there exists an INS $\bar{p}$ such that $\bar{p}(s'|s,a) = \prod_{i=1}^{n} p_i(s_i'|s_{N_i^\kappa}, a_i)$. An environment is ξ-dependent, if:*

$$\sup_{s,a} D_{\mathrm{TV}}(\, p(s'|s,a) \parallel \bar{p}(s'|s,a)) = \sup_{s,a} \frac{1}{2} \sum_{s' \in \mathcal{S}} |p(s'|s,a) - \bar{p}(s'|s,a)| \leq \xi.$$

Denote $\hat{p}(s'|s,a) = \prod_{i=1}^{n} \hat{p}_i(s_i'|s_{N_i^\kappa}, a_i)$. By using $\hat{p}(s'|s,a)$ to approximate the true dynamics, larger κ leads to better approximation of model p, but also heavier computation overhead. The universal model error $D(\, p \parallel \hat{p})$ can be divided into two parts: dependency bias $D(\, p \parallel \bar{p})$ and independence-approximation error $D(\, \bar{p} \parallel \hat{p})$ with

$$D(\, p \parallel \hat{p}) \leq D(\, p \parallel \bar{p}) + D(\, \bar{p} \parallel \hat{p}). \tag{12}$$

More details of networked MDP and distances among $\bar{p}, \hat{p}$ and $\hat{p}$ are shown in Fig. 2b–d. Then for an ξ-dependent system, when the model becomes very accurate, meaning $D(\, \bar{p} \parallel \hat{p}) \approx 0$, $\sup D(\, p \parallel \hat{p}) \approx \sup D(\, p \parallel \bar{p}) = \xi$. While D can be any appropriate distance metric, we use the TV distance hereafter for the ease of presentation. Here we give analysis under both independent and ξ-dependent networked systems. In the following experiments, the systems used to evaluate the algorithms also have the properties of an ξ-dependent system.

Let π indicate a collective policy $\pi = [\pi_1, \ldots, \pi_n]$, and the model $\hat{p}$ be an INS $\hat{p}(s'|s,a) = \prod_{i=1}^{n} \hat{p}_i(s_i'|s_{N_i}, a_i)$ that approximating the true MDP. Denote the data-collecting policy as π_{D}. For decentralized learning and control, each agent learns a localized model $\hat{p}_i$, policy $\pi_i(\cdot|s_{N_i})$. For each agent i, we solve the following problem:

$$\pi_i^{k+1}, p_i^{k+1} = \underset{\pi_i, p_i}{\operatorname{argmax}} \quad \hat{\eta}[\pi] - C(\, p, \hat{p}, \pi, \pi_{\mathrm{D}}). \tag{13}$$

Below, we first lay out the monotonic improvement property of independent and ξ-independent systems. Then we describe our framework in Algorithm 1, and Fig. 2e shows the model learning process. Experience data of state transition and rewards from the interaction between the environment and the agent are used to train the environment model. Then our method can use the large amount of data generated by the interaction between the model and the agent to improve the sampling efficiency in the training process.

Monotonic model-based improvement

In model-based learning, different rollout schemes can be chosen. The vanilla rollout assumes that models are used in an infinite horizon. The branched rollout performs a rollout from a state sampled by a state distribution of previous policy π_{D}, and runs T steps in $\hat{\pi}$ according to π. Based on different rollout schemes, we can construct two lower bounds. Under vanilla rollout, real return and model return can be bounded by model error and policy divergence. Formal results are presented in Theorem 1. The detailed proof is deferred to Supplementary Section 4.2.

Theorem 1. *Consider an INS. Denote local model errors as:*

$$\epsilon_{m_i} = \max_{s_{N_i}, a_i} D_{\mathrm{TV}}[\, p_i(s_i'|s_{N_i}, a_i) \parallel \hat{p}_i(s_i'|s_{N_i}, a_i)]$$

and divergences between the data-collecting policy and evaluated policy as:

$$\epsilon_{\pi_i} = \max_{s_{N_i}} \mathrm{D}_{TV}[\pi_D(a_i|s_{N_i}) \parallel \pi(a_i|s_{N_i})]$$

Assume the upper bound of rewards of all agents is $r_{\max}$. Let $\eta^p[\pi_1, \ldots, \pi_n]$ denote the real returns in the environment. Also, let $\eta^{\hat{p}}[\pi_1, \ldots, \pi_n]$ denote the returns estimated in the model trajectories, and the states and actions are collected with π_{D}. Then we have:

$$|\eta^p[\pi_1, \ldots, \pi_n] - \eta^{\hat{p}}[\pi_1, \ldots, \pi_n]| \leq \frac{2 r_{\max}}{1-\gamma} \sum_{i=1}^{n} \left[\frac{\epsilon_{\pi_i}}{n} + (\epsilon_{m_i} + 2\epsilon_{\pi_i}) \sum_{k=0}^{\infty} \gamma^{k+1} \frac{|N_i^k|}{n} \right]. \tag{14}$$

Intuitively, the term $\sum_{k=0}^{\infty} \gamma^{k+1} \frac{|N_i^k|}{n}$ would be in the same magnitude as $\frac{1}{1-\gamma}$, which might be huge given the choice of γ, making the bound too loose to be effective. To make tighter the discrepancy bound in Theorem 1, we adopt the branched rollout scheme. The branched rollout enables a effective combination of model-based and model-free rollouts. For each rollout, we begin from a state sample from $d_{\pi_{\mathrm{D}}}$, and run T steps in each localized $\hat{\pi}_i$. When branched rollout is applied in an INS, Theorem 2 gives the returns bound.

Theorem 2. *Consider an INS. Denote local model errors as:*

$$\epsilon_{m_i} = \max_{s_{N_i}, a_i} D_{\mathrm{TV}}[\, p_i(s_i'|s_{N_i}, a_i) \parallel \hat{p}_i(s_i'|s_{N_i}, a_i)]$$

and divergences between the data-collecting policy and evaluated policy as:

$$\epsilon_{\pi_i} = \max_{s_{N_i}} \mathrm{D}_{TV}[\pi_D(a_i|s_{N_i}) \parallel \pi(a_i|s_{N_i})]$$

Assume the upper bound of rewards of all agents is $r_{\max}$. Let $\eta^p[\pi_1, \ldots, \pi_n]$ denote the real returns in the environment. Also, let $\eta^{\text{branch}}[\pi_1, \ldots, \pi_n]$ denote the returns estimated via T-step branched rollout scheme. Then we have:

$$|\eta^p[\pi_1, \ldots, \pi_n] - \eta^{\text{branch}}[\pi_1, \ldots, \pi_n]| \leq \frac{2r_{\max}}{1-\gamma} \sum_{i=1}^{n} \left[\epsilon_{m_i} \cdot \left(\sum_{k=0}^{T-1} \gamma^{k+1} \frac{|N_i^k|}{n} \right) + \epsilon_{\pi_i} \cdot \left(\sum_{k=T}^{\infty} \gamma^{k+1} \frac{|N_i^k|}{n} \right) \right]$$

Comparing the results in Theorems 1 and 2, we can see that branched rollout scheme reduced the coefficient before ϵ_{m_i} from $\sum_{k=0}^{\infty} \gamma^{k+1} \frac{|N_i^k|}{n} \leq \frac{\gamma}{1-\gamma}$ to $\sum_{k=0}^{T-1} \gamma^{k+1} \frac{|N_i^k|}{n} \leq \sum_{k=0}^{T-1} \gamma^{k+1} = \frac{\gamma(1-\gamma^T)}{1-\gamma}$. This reduction explains that empirically, branched rollout brings better asymptotic performance. Also, if we set $T = 0$, this bound turn into a model-free bound. This indicates that when ϵ_{m_i} is lower than ϵ_{π_i} allowed by our algorithm, a model might increase the performance.

Algorithm 1: A general model-based learning framework.

1: Initialize policy $\pi(a|s) = \{\pi_i\}_{i=1}^n$, predictive model $p_\psi(s'|s,a) = \{p_{\psi_i}(s_i|s_{N_i^\kappa}, a_i)\}_{i=1}^n$, empty dataset $\{\mathcal{D}_i\}_{i=1}^n$.
2: **for** N epochs **do**
3: Collect data with $\{\pi_i\}_{i=1}^n$ in real environment: $\mathcal{D}_i = \mathcal{D}_i \cup \left\{\left(s_i^t, a_i^t, s_i^{t+1}, r_i^t\right)\right\}_t$ for all agents
4: Train model p_{ψ_i} on dataset $\mathcal{D}_i$ via maximum likelihood: $\psi_i \leftarrow \text{argmax}_{\psi_i} \mathbb{E}_{\mathcal{D}_i}\left[\log p_{\psi_i}(s_i|s_{N_i^\kappa}, a_i)\right]$
5: Optimize policy under predictive model: $\pi_i^{k+1} \leftarrow \text{argmax}_{\pi_i'} \hat{\eta}[\pi'] - C(p, \hat{p}, \pi', \pi_D)$.
6: **end for**

Incorporating dependency bias

In reality, not every system satisfies the definition of an INS. Yet we can generalize Theorem 2 into an ξ-dependent system.

Corollary 1. *Consider an ξ-dependent networked system. Denote local model errors as:*

$$\epsilon_{m_i} = \max_{s_{N_i}, a_i} D_{\text{TV}}[\, p_i(s_i'|s_{N_i}, a_i) \parallel \hat{p}_i(s_i'|s_{N_i}, a_i)]$$

and divergences between the data-collecting policy and evaluated policy as:

$$\epsilon_{\pi_i} = \max_{s_{N_i}} \mathrm{D}_{TV}[\pi_D(a_i|s_{N_i}) \parallel \pi(a_i|s_{N_i})]$$

Assume the upper bound of rewards of all agents is $r_{\max}$. Let $\eta^p[\pi_1, \ldots, \pi_n]$ denote the real returns in the environment. Also, let $\eta^{\text{branch}}[\pi_1, \ldots, \pi_n]$ denote the returns estimated via T-step branched rollout scheme. Then we have:

$$|\eta^p[\pi_1, \ldots, \pi_n] - \eta^{\text{branch}}[\pi_1, \ldots, \pi_n]| \leq \frac{2r_{\max}\gamma}{(1-\gamma)^2}\xi + \frac{2r_{\max}}{1-\gamma} \sum_{i=1}^{n} \left[\epsilon_{m_i} \cdot \left(\sum_{k=0}^{T-1} \gamma^{k+1} \frac{|N_i^k|}{n} \right) + \epsilon_{\pi_i} \cdot \left(\sum_{k=T}^{\infty} \gamma^{k+1} \frac{|N_i^k|}{n} \right) \right]$$

The proof can also be found in Supplementary Information. Compared with Theorem 2, Corollary 1 is more general, as it applies to multi-agent systems that are not fully independent. Intuitively, if a networked system seems nearly independent, local models will be effective enough. The bound indicates that when the policy is optimized in a trust region where $D(\pi, \pi_D) \leq \epsilon_{\pi_i}$, the bound would also be restricted, making monotonic update more achievable.

Algorithm 2: Model-based decentralized policy optimization.

1: Initialize the model buffer $\mathcal{D}_i^{\text{M}}$ and the environment buffer $\mathcal{D}_i^{\text{E}}$, where $i \in \mathcal{V}$.
2: Initialize the rollout length T and adjacency matrix A through κ.
3: Initialize the decentralized model p^{ψ_i}, actor π^{θ_i} and critic V^{ϕ_i} by adjacency matrix A, where $i \in \mathcal{V}$.
4: **for** many epochs **do**
5: Sample action according to decentralized π^{θ_i} and local observation $s_{N_i^\kappa}$ in environment to collect trajectories τ_i^{E}.
6: Store trajectories to environment buffer $\mathcal{D}_i^{\text{E}}$ for each agent i, $\mathcal{D}_i^{\text{E}} = \mathcal{D}_i^{\text{E}} \cup \{\tau_i^{\text{E}}\}$.
7: Minimize the objective in equation (6) to update model p^{ψ_i} through trajectories from $\mathcal{D}_i^{\text{E}}$.
8: **for** many steps **do**
9: **for** many rollouts **do**
10: Sample s_i^t from $\mathcal{D}_i^{\text{E}}$ as an initial state.
11: Simulate T-step trajectories τ_i^{M} from s_i^t by decentralized policy π^{θ_i} and decentralized model p^{ψ_i}.
12: Store trajectories τ_i^{M} to model buffer $\mathcal{D}_i^{\text{M}}$ for each agent i, $\mathcal{D}_i^{\text{M}} = \mathcal{D}_i^{\text{M}} \cup \{\tau_i^{\text{M}}\}$.
13: **end for**
14: **for** many gradient steps **do**
15: Update the extended value functions V^{ϕ_i} on trajectories τ_i^{M} sampled from $\mathcal{D}_i^{\text{M}}$ according to equation (3)
16: Update the decentralized policies π^{θ_i} on trajectories τ_i^{M} sampled from $\mathcal{D}_i^{\text{M}}$ according to equation (5).
17: **end for**
18: **end for**
19: **end for**

Compared with the theorems of model-based policy optimization in the single-agent domain[54], we extend them to the multi-agent domain, especially for completely independent networked systems or even more general and real-world systems named ξ-dependent networked systems. Further, to solve the problems of the increase of computation complexity, communication cost and system instability caused by the multi-agent system, we propose a novel communication mechanism and analyse the accuracy of the estimation of value functions and policy gradients through the following theoretical proofs, which is completely different from the general model-based algorithms.

Therefore, the challenges for our algorithm and theorems are mainly twofold in these ξ-dependent networked systems with fewer communications. One is that local communication and approximation dependency bias $D(p \parallel \bar{p})$ between independent networked systems and ξ-dependent networked systems will increase the generalization error ϵ_{m_i} of the environment model and policy shift ϵ_{π_i} between π and π_D, which will further increase the lower bounds $C(p, \hat{p}, \pi, \pi_D)$ in Theorems 1 and 2. The results of experiments fully prove that our method can reduce the influence of these errors on the monotonic improvement of the model-based learning through the following algorithm design and hyperparameter selection. The other is that our method needs to provide accurate policy gradients and value functions for the multi-agent system to guide the policy optimization, which is explained in more detail in Theorems 3 and 4.

Analysis of policy gradients approximation

In this section, we show that the policy gradient induced by equation (5) is an asymptotic approximation of the true policy gradients. We begin with discussing that the extended value function $V_i(s_{N_i^\kappa})$ is a good approximation of the real value function. We formally state the result in Theorem 3 and defer the proof to Supplementary Section 4.5.

Theorem 3. *Define* $V_i(s_{N_i^\kappa}) = \mathbb{E}_{s_{N_{-i}^\kappa}}\left[\sum_{t=0}^{\infty} r_i^t(s_t, a_t) | s_{N_i^\kappa}^0 = s_{N_i^\kappa}\right]$ *and* $V_i(s) = \mathbb{E}[\sum_{t=0}^{\infty} r_i^t | s^0 = s]$, *then*

$$|V_i(s) - V_i(s_{N_i^\kappa})| \leq \frac{r_{\max}}{1-\gamma}\gamma^\kappa. \tag{15}$$

Remark 1. *Recall that* $V(s) = \frac{1}{n}\sum_{i=1}^{n} V_i(s)$. *From equation* (15), *it is easy to obtain the following result*

$$|V(s) - \frac{1}{n}\sum_{i=1}^{n} V_i(s_{N_i^\kappa})| \leq \frac{r_{\max}}{1-\gamma}\gamma^\kappa, \tag{16}$$

which indicates that the global value function $V(s)$ *can be approximated by the average of all extended value functions.*

In policy optimization, value functions are used for calculating advantages $\hat{A}^{(t)}$, and we have shown that $V(s)$ can be estimated with the average of extended value functions $\frac{1}{n}\sum_{i=1}^{n} V_i(s_{N_i^\kappa})$. In practice, an agent might not get the value function of distant agents and can access only the value function of its κ-hop neighbours. However, we can prove that $\tilde{V}_i = \frac{1}{n}\sum_{j\in N_i^\kappa} V_j(s_{N_j^\kappa})$is already very accurate for calculating the policy gradient for agent i. This theoretical result has been extended to general multi-agent systems based on previous research[39]. Theorem 4 formally states this result and the proof is deferred to Supplementary Section 4.6.

Theorem 4. *Let* $\hat{A}_t = r^{(t)} + \gamma V(s^{(t+1)}) - V(s^{(t)})$ *be the TD (temporal difference) residual, and* $g_i = \mathbb{E}[\hat{A}\nabla_{\theta_i}\log\pi_i(a|s)]$*be the policy gradient. If* $\tilde{A}_t$ and $\tilde{g}_i$ *are the TD residual and policy gradient when value function* $V(s)$ *is replaced by* $\tilde{V}_i(s) = \frac{1}{n}\sum_{j\in N_i^\kappa} V_j(s_{N_i^\kappa})$, *we have:*

$$|g_i - \tilde{g}_i| \leq \frac{\gamma^{\kappa-1}}{1-\gamma}\left[1 - (1-\gamma^2)\frac{N_i^\kappa}{n}\right] r_{\max} g_{\max}, \tag{17}$$

where $r_{\max}$*and* $g_{\max}$ *denote the upper bound of the absolute value of reward and gradient, respectively.*

Remark 2. *Theorem 4 justifies that the policy gradients computed based on the sum of the neighbouring extended value functions is a close approximation of true policy gradients. The power of this theorem is that the extended value function* $V_i(s_{N_i}^\kappa)$ *requires only the neighbouring information, thus easier to approximate and scalable. Despite the reduction in computation, the difference between the approximated and true gradient in equation* (17) *is small.*

On the basis of the difficulties and challenges proposed in the above theoretical analysis, we designed corresponding solutions in the algorithm and the specific implementation details can be found in our code. In Theorems 1 and 2, the probability to monotonic improvement of the model-based learning in the algorithm will be increase when the lower bound $C(p, \hat{p}, \pi, \pi_D)$that consists of two parts: generalization model error ϵ_{m_i} and policy shift ϵ_{π_i} is tight enough to constrain equation (10). To reduce the generalization model error ϵ_{m_i} caused by dependency bias $D(p \parallel \hat{p})$in equation (12), we design a stage of warm-up in our algorithm to pre-train the model until an acceptable accuracy of the model is obtained and a method to select data from the true MDP or model with a probability to guide the usage of the model, and we further adopted a k-branched rollout method to improve the accuracy of model. According to equation (12), when the model becomes very accurate, meaning $D(\bar{p} \parallel \hat{p}) \approx 0$, $\sup D(p \parallel \hat{p}) \approx \sup D(p \parallel \bar{p}) = \xi$. Then we design a distribution of the neighbours for each agent, including the quantity and topology of neighbours, to reduce the model error ϵ_{m_i} caused by dependency bias $D(p \parallel \bar{p})$ in the experiments. Lastly, to constrain the policy shift ϵ_{π_i} between π and π_D, we use the method of PPO to limit the difference in policies.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The SUMO data generated during other experimental processes, as well as the power-grid and pandemic data used, are all included in the code repository and can be used directly https://github.com/CDM1619/Networked-MB-MARL (ref. 83). The specific usage methods are further explained in README.md. The real data of Real Power-Net are available in https://drive.google.com/file/d/1-GGPBSolVjX1HseJVblNY3KoTqf-blmLh/view. This is a publicly available dataset from previous research work[73]. For more details on how to use this data to run the experiments, please refer to https://github.com/CDM1619/Networked-MB-MARL. Source data are provided with this paper.

Code availability

An implementation of our method is available at https://github.com/CDM1619/Networked-MB-MARL. We provide detailed installation tutorials and running examples in https://github.com/CDM1619/Networked-MB-MARL (ref. 83).

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Author contributions

C.M., A.L., Y.D., H.D. and Y.Y. designed the project. C.M., A.L., Y.D. and Y.Y. conducted the experiments and analysed the results. C.M., A.L., Y.D., H.D. and Y.Y. wrote the paper and response letter. C.M., A.L., Y.D. and Y.Y. analysed the theoretical results. C.M., A.L., Y.D., H.D. and Y.Y. participated in the discussion of the revised content.

Competing interests

The authors declare no competing interests.

Additional information

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Data collection	Part of the data in our study was generated through the simulator provided by the code: https://github.com/CDM1619/Networked-MB-MARL Another part of the data source has been provided with specific links in the "Data availability" section and supplementary information in the manuscript, The real data of Power-Net are available in https://drive.google.com/file/d/1-GGPBSolVjX1HseJVblNY3KoTqfblmLh/view
Data analysis	All data analysis in our study used the most widely recognized analysis method in the Machine/Reinforcement Learning community, without any special analysis methods. The specific analysis methods have been elaborated in detail in the manuscript. The basic programs for data drawing and analysis are also included in the code repository: https://github.com/CDM1619/Networked-MB-MARL

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The SUMO data generated during other experimental processes, as well as the power-grid and pandemic data used, are all included in the code repository and can be used directly \url{https://github.com/CDM1619/Networked-MB-MARL}. The specific usage methods are further explained in README.md. The real data of Real Power-Net are available in \url{https://drive.google.com/file/d/1-GGPBSolVjX1HseJVblNY3KoTqfblmLh/view}, This is a publicly available dataset from previous research work. For more details on how to use this data to run the experiments, please refer to \url{https://github.com/CDM1619/Networked-MB-MARL}.

Human research participants

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Supplementary information

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Efficient and scalable reinforcement learning for large-scale network control

In the format provided by the authors and unedited

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1 Comparison with Existing Work

We have comprehensively analyzed the relevant work in the field and added more baselines. Our baselines consist of two main categories: model-based algorithms such as MAG[1] and model-free algorithms. In addition, we add some algorithms which are the ablation of our method on different components such as extended value functions, adjacent information, and local models. Next, we will describe the existing work and ablation separately.

1.1 Model-based learning in multi-agent systems

In the work about model-based multi-agent reinforcement learning, we focused on analyzing the following works and compared MAG as one of the baselines through experiments.

- MAG[1] , a multi-agent model optimization framework, adopts a novel approach whereby local models are conceptualized as multi-step decision-making agents, and the current policies are viewed as the dynamics shaping the model rollout process. This unique perspective enables the local models to incorporate the intricate interplay and mutual influence across multiple time steps, offering a more holistic consideration before generating predictions. But MAG uses global information that has been processed by the Attention block as the input of the critic. This is still essentially a centralized training method, only using local models, while in our method, the models, polices, and critics are all decentralized.
- Furthermore, some exist model-based multi-agent reinforcement learning endeavors that concentrate on goal alignment within the context of navigation ensemble tasks[2]. These works presuppose the availability of pre-trained peer-to-peer navigation policies and models. They construct a modular system composed of learning components. However, they do not delve into the specifics of model learning theory or policy improvement theory.
- In certain model-based multi-agent reinforcement learning endeavors, communication and interaction mechanisms are also employed[3]. Agents engage in iterative exchanges of their individual forecasts to collaboratively plan ahead. Each agent has access to a learned world model capable of generating future state and reward predictions conditioned on actions sampled from the agent's policy. Subsequently, these insights are encoded into messages and employed in training through differentiable message passing to learn communication protocols. However, this approach does not take into account scenarios with constrained communication, rendering it unsuitable for large-scale real-world systems subject to objective communication restrictions. And there is no theoretical joint analysis of model learning and strategy learning errors.
- Some researchers have also proposed an Intention Sharing (IS) mechanism[4] to enhance the coordination among agents. In this scheme, each agent employs an attention mechanism based on messages received from other agents to learn the relative significance of components within imagined trajectories. This compression allows for capturing the agent's future action plans in the form of an intent message for communication. Nevertheless, this method still disregards communication limitations prevalent in the real world. Additionally, it necessitates extra information learning, thereby introducing additional computational costs.

In a comprehensive assessment, the existing model-based approaches for multi-agent reinforcement learning suffer from the following issues: (1) Despite being rooted in local models, they still rely on a global critic, rendering them ineffective in communication-constrained scenarios. (2) Theoretical analysis and bounded estimation for multi-agent model learning are lacking, resulting in an insufficient grasp of the model's impact on reinforcement learning. (3) The integration of model learning and policy learning, a potential iterative process in real-world scenarios, has not been achieved.

Supplementary Table 1. Comparison of Local Message Sizes (LMS) of different decentralized algorithms.

	DIAL[5], NeurComm[6]	CommNet[7]	FPrint[8]	ConseNet[9]	**Ours**
local message sizes	$O(\|s_i\|+\|\pi_i\|+\|h_i\|)$	$O(\|s_i\|+\|h_i\|)$	$O(\|s_i\|+\|\pi_i\|)$	$O(\|s_i\|)$	$O(\|s_i\|)$

1.2 Model-free learning and communication mechanism

This categories are further subdivided into hidden information-based algorithms DIAL[5], CommNet[7], NeurComm[6], IC3Net[10] and non-hidden information-based algorithms ConseNet[9], FPrint[8]. Note that hidden information-based algorithms necessitate a larger number of messages to be transmitted, consequently demanding increased communication bandwidth and imposing a heavier computational burden. We define the $h_{i,t}$[6]

$$h_{i,t} = g_{v_i}\left(h_{i,t-1}, e_{\lambda_i^s}\left(s_{\gamma_{i,t}}\right), e_{\lambda_i^\pi}\left(\pi_{N_i,t-1}\right), e_{\lambda_i^h}\left(h_{N_i,t-1}\right)\right) \tag{1}$$

where $h_{i,t}$ is the hidden state (or the belief) of each agent i in time t and e_{λ_i} and g_{v_i} are differentiable message encoding and extracting functions. In Table 1, we present the local message sizes[6] for different algorithms, it represents the amount of information required for each local agent when learning and reasoning, and it is evident that our approach utilizes the minimal amount of information.

- **Hidden information-based algorithms** The DIAL approach distinguishes itself by encoding received messages individually instead of resorting to averaging, although the summation of all encoded inputs remains unchanged. In contrast, CommNet adopts a more lossy aggregation mechanism which involves averaging the received messages prior to encoding, followed by the summation of all encoded inputs. Moreover, both DIAL and CommNet eschew the inclusion of policy fingerprints within their respective messages. Within NeurComm, each agent effectively leverages delayed global information to refine its belief system, while also acquiring the capacity to learn the messages to optimize the control performance of all other agents. Learning hidden information requires additional computational burden and may even increase decision errors. Certain studies have directed their attention towards communication strategies that involve the selective transmission of messages. For instance, ATOC[11] employs a soft attention mechanism to apportion a communication probability to individual agents, whereas IC3Net[10] adopts a hard binary attention approach to determine whether communication should occur. In IC3Net, the agents maintain their local hidden states with a LSTM kernel. Such endeavors prove particularly beneficial when each agent is compelled to establish communication priorities. It is important to note that these approaches find heightened relevance in scenarios where the necessity for communication target prioritization is prominent. However, in the context of Scenarios with limited communication, this need is less pronounced, as the communication scope is typically confined to localized neighborhoods. This method still uses global information.

- **Non-hidden information-based algorithms** ConseNet and FPrint exclusively rely on neighborhood information. In the case of ConseNet, an additional consensus update mechanism is employed, facilitating the overwrite of critic parameters with the mean values obtained from all critics within the enclosed neighborhood. On the other hand, FPrint incorporates the inclusion of neighbor policies as an integral component of its framework. It is worth noting that in real scenarios, the difficulty for intelligent agents to obtain their neighbor policies is often larger than that of obtaining neighbor states through simple sensors. Our algorithm also does not require neighbor policy information.

1.3 Other ablation baselines

The following algorithm is the ablation of our method on different components such as extended value functions, adjacent information, and local models.

- CPPO [12,13]: Centralized PPO learns a centralized critic $V_i(s)$. This baseline aims to analyze the performance when κ is set to be arbitrarily huge, and is used in[13] as a benchmark algorithm for networked system control. The comparison between the performance of CPPO and our method can be viewed as an ablation study of extended value function and model usage.

- DPPO [12]: Decentralized PPO learns an independent actor and critic for each agent. We implement it by using the neighbor's state for extended value estimation. The comparison between the performance of DPPO and our method can be viewed as an ablation study of model usage.

- IMPO: A decentralized and model-based algorithm based on our method, but without information of neighbors and does not use the extended value function to predict global value function and gradient. The comparison between the performance of IMPO and our method can be viewed as an ablation study of the extended value function.

- MPO$_{na}$: A decentralized and model-based algorithm based on our method, but without information of non-adjacent neighbors. The comparison between the performance of MPO$_{na}$ and our method can be viewed as an ablation study of adjacent neighbors. The more detailed description is shown in **Fig. 1** in the main text.

1.4 Theoretical connections with previous work

In addition to the aforementioned aspects, prior work has introduced the foundational assumption of completely independent systems[14]. However, we believe that this assumption is considerably strong and not well-suited for the intricacies of real-world complex systems. Algorithms designed under such an assumption lack scalability, rendering them less viable for large-scale and intricate systems. As the number of agents grows, the approximation errors introduced by this assumption become challenging to control. Furthermore, due to the absence of model learning, these algorithms exhibit low exploration efficiency and sample efficiency in complex tasks. Rapidly attaining optimal policies becomes arduous. To address these limitations, we have

introduced the concept of ξ-dependent systems, which mitigates the issues associated with the independent systems assumption. Specifically, our basic theoretical starting point is from **Definition 1** to **Definition 2** in our manuscript.

Furthermore, we have incorporated model learning theory to tackle the challenges of sample inefficiency. Furthermore, we have incorporated model learning theory to tackle the challenges of sample inefficiency. The theoretical development of this part is based on the MBPO[15] of model-based single agent reinforcement learning. We extend model learning theory to the field of multi-agent systems with complex coupling relationships between agents and provide a theoretical guarantee for monotonic improvement of policies in multi-agent learning. Specifically, our basic theoretical starting point is from

$$\eta^{p}[\pi] \geq \eta^{\hat{p}}[\pi] - C^{15} \tag{2}$$

to

$$\eta^{p}[\pi_1, ..., \pi_n] \geq \eta^{\hat{p}}[\pi_1, ..., \pi_n] - C \tag{3}$$

In summary, our approach has extended the realm of networked multi-agent reinforcement learning to encompass larger scales and a wider array of intricate tasks. Although our theoretical findings are based on the work of Scalable SAC[14], we have made significant theoretical contributions to the practical application of the algorithm. To the best of our knowledge, our theoretical findings may represent the first provable assurance of policy monotonic improvement based on decentralized model learning, policy learning, and value function learning within scalable multi-agent networked systems. Additionally, our work stands as the first endeavor to extend rigorously independent multi-agent network systems to the realm of generalized multi-agent systems. Moreover, we introduce the first practically effective multi-agent reinforcement learning algorithm tailored for large-scale systems.

In conclusion, we have conducted a comprehensive analysis of the disparities between our work and existing research, delineating the advantages inherent to our approach. Notably, we have incorporated select aspects of these prior works as baselines in our experiments, including both model-based methods like MAG, and model-free algorithms such as NeurComm, CommNet, DIAL, IC3Net, FPrint, and ConseNet. The empirical results undeniably demonstrate the superiority of our algorithm over the existing methodologies, achieving optimal performance with minimal communication costs. Moreover, our contribution extends beyond empirical results; we provide foundational model learning theory and policy learning theory for model-based multi-agent reinforcement learning, enriching the theoretical underpinnings of the field.

2 Description of Systems

We described the real-world systems in our experimental scenario in Table 2, including their scale and real-world impact of the problem.

Supplementary Table 2. Summary of the scale of the problems we addressed and the sources of authenticity for the systems.

	System scale	Real-world impact
ATSC-New York	New York is a large city, with a total land area of approximately 1214 square kilometers. As of December 2023, the city's population is estimated at around 8340,000 individuals. We examined the performance at 436 different traffic intersections.	The ATSC New York system is developed based on the SUMO platform and real road network map of New York City. 1) SUMO was used to provide traffic forecasts for authorities of the City of Cologne during the Pope's visit in 2005 and during the Soccer World Cup 2006: . 2) SUMO is widely used by the V2X community for both, providing realistic vehicle traces, and for evaluating applications in an online loop with a network simulator.
Real Power-Net	This large-scale power system comprises over 400 photovoltaic (PV) nodes, which have been subdivided into 199 control zones. The grid's coverage spans approximately 300 square kilometers, equivalent to supporting the energy supply needs of a city like Philadelphia in the United States.	The load profiles[16] for each network are adjusted using real-time Portuguese electricity consumption data, which encompasses 232 consumers over a period of 3 years: . To account for varying solar radiation levels across different zones, we consider ten cities, regions, and provinces within Belgium, Netherlands, and Luxembourg. These include Antwerp, Brussels, Flemish-Brabant, Hainaut, Liege, Limburg, Luxembourg, Namur, Walloon-Brabant, and West-Flanders. The PV data is sourced from Elia Group: , the power network operator in Belgium.
Pandemic network	We optimized the virus containment strategies within distinct social settings with populations ranging from 500 to 10,000 people. These social settings encompassed a diverse array of establishments, including Homes, Grocery Stores, Offices, Schools, Retail Stores, Hospitals, Hair Salons, Restaurants, and Bars, each of which plays a crucial role in the transmission dynamics of the virus[17].	The population's age distribution is based on actual US population age distribution statistics. Hospital capacities in various societal segments are determined following Red Cross guidelines for hospital infrastructure and aligned with 2020 data from the American Hospital Association. Average virus infection and transmission rates, along with their standard deviations, are calibrated using COVID-19 hospitalization and mortality data from Sweden . Population mask-wearing data is sourced from the Health and Safety Fund of North America.

Flow This task consists of Figure Eight and Ring Attenuation. The objective is to achieve a target speed and avoid collision, which is similar to CACC. The road network has a shape of ring or figure "eight". The figure eight network, previously presented in[18], acts as a closed representation of an intersection. In a figure eight network containing a total of 14 vehicles, we witness the formation of queues resulting from vehicles arriving simultaneously at the intersection and slowing down to obey right-of-way rules. The state consists of velocity and position for the vehicle. The action is the acceleration of the vehicle $a \in \mathbb{R}_{[a_{\min}, a_{\max}]}$. The objective of the learning agent is to achieve high speeds while penalizing collisions. Accordingly, the local reward function is defined as $r_i = v_{des} - |v_{des} - v_i|$, where v_{des} is an arbitrary target velocity. In Ring Attenuation, the state and action of each vehicle is the same as Figure Eight. In the perspective of a networked system, we assume that the vehicles are connected with the preceding and succeeding vehicle, thus resulting in a loop-structured graph.

Cooperative Adaptive Cruise Control CACC consists of two scenarios: Catch-up and Slow-down. The objective of CACC is to adaptively coordinate a platoon of 8 vehicles to minimize the car-following headway and speed perturbations based on real-time vehicle-to-vehicle communication. The state of each agent consists of headway h, velocity v, acceleration a , and

is shared to neighbors within two steps. The action of each agent is to choose appropriate hyper-parameters $(\alpha^\circ, \beta^\circ)$ for each OVM controller[19], selected from four levels {(0,0),(0.5,0),(0,0.5),(0.5,0.5)}, where $\alpha^\circ, \beta^\circ$ denotes headway gain and relative gain. The reward is defined as $(h_{i,t} - h^*)^2 + (v_{i,t} - v_t^*)^2 + 0.1a_{i,t}^2$, where the target headway and velocity profile are $h^* = 20m$ and v_t^*. Whenever a collision happens ($h_{i,t} < 1m$), a large penalty of -1000 is assigned to each agent and the state becomes absorbing. In catch-up scenario, initial headway of the first vehicle is larger than the target headway, thus the following agents learn how to catch up with the first vehicle, where target speed $v_t^* = 15m/s$ and initial headway $h_{1,0} > h_{i,0}, \forall i \neq 1$. In slow-down scenario, target speed v_t^* linearly decreases to 15m/s during the first 30s and then stays at constant, thus agents learn how to slow down speed cooperatively,where initial headway $h_{i,0} = h^*$.

Adaptive Traffic Signal Control The objective of ATSC is to adaptively adjust signal phases to minimize traffic congestion based on real-time road-traffic measurements. Here we use two scenarios: ATSC Grid and ATSC Monaco. Both scenarios simulate peak-hour traffic. The MDP horizon is 720 steps, and one MDP step corresponds to a 5s simulation to prevent the traffic lights from switching too frequently. For safety, a 2 second yellow time is inserted before switching to red. **Figure 2** (d) in manuscript illustrates a 5×5 synthetic traffic grid: ATSC Grid traffic network, which is formed by a two-lane main road with a speed limit of 20m/s and a single-lane avenue with a speed limit of 11m/s. In the 5×5 synthetic traffic grid, the peak hour traffic dynamics are simulated by a collection of four time-variant traffic flows, including loading and recovery phases. The grid is homogeneous. The state of each agent is all the twelve flows, and all agents have the same action space, which is a set of five pre-defined signal phases: E-W straight phase, E-W left-turn phase, and three straight and left-turn phases for E, W, and N-S.. **Figure 2** (e) in manuscript illustrates a more challenging scenario with a heterogeneous network structure with diverse action and observation spaces: ATSC Monaco traffic network with 28 intersections. Four collections of traffic flows that are multiples of 325 veh/hr will be continuously generated to simulate peak-hour traffic. The blue intersections are different controlled objects with size of the action space changing from 2 to 6 and the dimension of observation space changing from 2 to 22. The map of the ATSC Monaco system is derived from the actual traffic grid of the city of Monaco, enabling a reasonable simulation of traffic conditions within a heterogeneous and asymmetric urban network. The ATSC Monaco system is developed based on the SUMO platform. SUMO has been used within several projects for answering a large variety of research questions and real applications:

- Evaluate the performance of traffic lights, including the evaluation of modern algorithms up to the evaluation of weekly timing plans.
- Vehicle route choice has been investigated, including the development of new methods, the evaluation of eco-aware routing based on pollutant emission, and investigations on network-wide influences of autonomous route choice.
- SUMO was used to provide traffic forecasts for authorities of the City of Cologne during the Pope's visit in 2005 and during the Soccer World Cup 2006 [1].
- SUMO is widely used by the V2X community for both, providing realistic vehicle traces, and for evaluating applications in an on-line loop with a network simulator.

More details can be checked in code and videos.

Decentralized Inverter-based Secondary Voltage Control To further demonstrate the capabilities of our algorithm, we apply the developed our method to more complex multi-agent Power-Grid systems, IEEE 34-bus test feeder with 20 and 40 heterogeneous distributed generators. Specifically, this is a decentralized inverter-based secondary voltage control problem in distributed generators, which can be formulated as a cooperative MARL problem. The objective of Power-Grid is to maintain the voltage and frequency of all distributed generators to the reference value when the disruptions in the systems and tracking imperfections in primary control. **Figure 2** (f) in manuscript illustrates Power-Grid systems. This simulator is based on the line and load specifications detailed in[20] and[21]. The action for each agent is the secondary voltage control set point. Specifically, we discretize the action space uniformly to a value between 1.00 pu and 1.14 pu in this system. The state of each agent i consists of 9 variables $s_{i,t} = (\delta_i, P_i, Q_i, i_{odi}, i_{oqi}, i_{bdi}, i_{bqi}, v_{bdi}, v_{bqi})$, where δ_i denote the measured reference angle frame; P_i and Q_i is the active and reactive power, respectively. i_{odi}, i_{oqi}, i_{bdi} and i_{bqi} denote the output d-q currents of the agent i and the directly connected buses, respectively; v_{bdi} and v_{bqi} represent the output d-q voltages of the connected bus, respectively; These definitions and descriptions refer to previous research[22]. In order to stabilize the voltage of the Power-Grid systems, we design a positive reward to encourage agents when the distributed generators with voltage close to 1.00 pu and design a negative reward to punish agents when the distributed generators with diverged voltages or have no power flow solution. Furthermore, to make the simulator closer to the real-world power systems, some random disturbances in the range of $\pm 5\%$ of the nominal values for each load are added in each simulation time step.

[1] https://eclipse.dev/sumo/about/

The Real-world Active Voltage Control To further validate the efficacy of our method in real-world scenarios, we tackle the challenge of active control in a large-scale power system[16] constructed with real-world data. The emerging trend of decarbonization is placing excessive stress on power distribution networks. Active voltage control is seen as a promising solution to relieve power congestion and improve voltage quality without extra hardware investment, taking advantage of the controllable apparatuses in the network, such as roof-top photovoltaics (PVs) and static var compensators (SVCs). These controllable apparatuses appear in a vast number and are distributed in a wide geographic area, making MARL a natural candidate. These power systems consist of 141 and 322 buses, respectively. To facilitate operational efficiency, a large-scale power network is partitioned into multiple regions, with each region being equipped with several photovoltaic (PV) installations overseen by the respective distribution network owner. Within this setup, each PV is equipped with an inverter that generates reactive power to regulate the voltage around a fixed reference value denoted as v_{ref}. In our task, we assume the independence and segregation of PV ownership (e.g., individual homeowners or enterprises) from distribution network ownership. Notably, the operation of PV infrastructure falls under the responsibility of the PV owner. This implies that each PV can be controlled in a distributed manner, naturally lending itself to the characterization of an agent. The objective is to ensure that the power systems is within safe control range and has minimal power losses. The load profiles of each electrical network are derived from real-time Portuguese electricity consumption data spanning a period of three years, encompassing 232 consumers. In order to highlight the distinctions between residential and industrial users, the default power factors specified in the case files are randomly perturbed by $\pm 5\%$, thereby generating corresponding real-time reactive power consumption. Solar data, sourced from Elia group, the power network operator in Belgium, is obtained and incorporated. Both the load and photovoltaic (PV) data are interpolated using a resolution of 3 minutes to align with the real-time control period within the power grid. To differentiate between varying solar radiation levels across different regions, the PV generation profiles from 10 cities/regions over a three-year period are collected, with PV systems within the same control region sharing identical generation profiles[16]. The specific real system situation is as follows:

- **Load Profiles** The load profiles for each network undergo modification based on real-time Portuguese electricity consumption data encompassing 232 consumers over a span of 3 years [2]. The initial dataset comprises electricity usage records of 370 residential and industrial clients between 2011 and 2014, captured at 15-minute intervals. Due to some data not commencing from the beginning, records are collected from January 1st, 2012, starting at 00:15:00. Locations featuring over 20 missing data entries are omitted, while the remaining gaps, primarily arising from the transition between winter time and daylight saving time, are linearly interpolated. Subsequently, the load data is interpolated at a 3-minute resolution to align with the real-time control intervals in the grid. The ultimate dataset size stands at 526080×232[16], encompassing load profiles for 232 consumers spanning 1096 days (equivalent to three years).

- **PV Profiles** To account for varying solar radiation levels across distinct zones, we consider ten cities, regions, and provinces within Belgium, Netherlands, and Luxembourg. These encompass Antwerp, Brussels, Flemish-Brabant (a province of Belgium), Hainaut (a province of Belgium), Liege, Limburg (a province of Netherlands), Luxembourg, Namur, Walloon-Brabant (a province of Belgium), and West-Flanders (a province of Belgium). The PV data is sourced from Elia Group [3], the power network operator in Belgium. This PV data is also subjected to interpolation at a 3-minute resolution, culminating in a total dataset size of 526080×10[16].

The active voltage control problem exhibits several intriguing characteristics. Firstly, it manifests as a convergence of both local and global factors. This denotes that while the voltage of each node is influenced by the real and reactive powers of all other nodes, the influence diminishes as the distance between nodes increases. Secondly, it constitutes a constrained optimization quandary, with the voltage threshold serving as the constraint and the minimization of total power loss standing as the objective. However, the relationship between the constraint and the control action lacks explicit definition. Thirdly, the structure of a distribution network adheres to a radial topology, offering substantial underlying information for control strategies. Nevertheless, the node-branch parameters of this topology can introduce significant uncertainties. Lastly, voltage control possesses a relatively considerable margin of tolerance, resulting in less severe consequences in cases where the control fails to meet standard requirements. This particular characteristic renders it highly suitable as a practical application scenario for multi-agent reinforcement learning that relies on a certain degree of exploration and trial-and-error. It serves as an ideal context for the implementation of grounded multi-agent reinforcement learning applications.

The load profiles of each electrical network are derived from real-time Portuguese electricity consumption data spanning a period of three years, encompassing 232 consumers. In order to highlight the distinctions between residential and industrial users, the default power factors specified in the case files are randomly perturbed by $\pm 5\%$, thereby generating corresponding real-time reactive power consumption. Solar data, sourced from Elia group, the power network operator in Belgium, is obtained

[2] https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014
[3] https://www.elia.be/en/grid-data/power-generation/solar-pv-power-generation-data

and incorporated. Both the load and photovoltaic (PV) data are interpolated using a resolution of 3 minutes to align with the real-time control period within the power grid. To differentiate between varying solar radiation levels across different regions, the PV generation profiles from 10 cities/regions over a three-year period are collected, with PV systems within the same control region sharing identical generation profiles[16].

Pandemic Networked Control for COVID-19 Mitigation[17] To further promote the applicability of our algorithm, we simulated prevention and control strategies that are close to real infectious diseases in large societies. The construction of this pandemic network relies on COVID-19 prevention and control policies and infection data by the real Swedish government. The year 2020 bore witness to the emergence of the COVID-19 virus, precipitating one of the most severe global pandemics in recorded history. Consequently, governments worldwide have been confronted with the arduous task of safeguarding public health while simultaneously striving to ensure the uninterrupted functioning of their respective economies. Epidemiological models serve as invaluable tools for comprehending the transmission dynamics of such diseases and forecasting the potential impact of intervention measures. Nevertheless, prevailing intervention policies, despite being rooted in data-driven approaches, still heavily rely on heuristic strategies. In this realistic scenario, we evaluate the utilization of our method in the optimization of mitigation policies aimed at minimizing the economic repercussions of a pandemic while ensuring the resilience of hospital capacity and reducing death cases. The difference between this pandemic simulator and traditional models is that it can capture complex interactions between individuals in specific locations within the community. We optimized the virus containment strategies within distinct social settings with populations ranging from 500 to 10,000 individuals. These social settings encompassed a diverse array of establishments, including Home, Grocery Store, Office, School, Retail Store, Hospital, Hair Salon, Restaurant, and Bar, each of which plays a crucial role in the transmission dynamics of the virus. By tailoring the strategies to the specific characteristics and interactions within these settings, we aimed to enhance the effectiveness of the virus control measures employed. The description sources for real data are as follows:

- The age distribution of the population is based on real US population age distribution statistics [4]
- Hospital capacities in various societal segments are determined based on the guidelines provided by the Red Cross for hospital infrastructure, aligned with the data from the American Hospital Association for the year 2020. The total bed count, distributed across 6,146 hospitals in the United States, amounts to 924,107 beds, with an average of approximately 150 beds per hospital.
- The average values and standard deviation of virus infection and transmission rates are calibrated based on the hospitalization and mortality data of COVID-19 cases in Sweden.
- The relevant data on the wearing of masks by the population comes from Health and Safety Fund of North America [5].
- In order to enhance the authenticity of our model, we conducted a calibration by comparing the average time to peak for deaths and tested critical cases within the simulator with actual data from Sweden, sourced from the World Health Organization [6]. We chose Sweden as our data reference due to its comparatively minimal restrictions during the initial pandemic wave. As a result, the virus dynamics observed in Sweden are regarded as more representative of natural trends. The main prevention and control policies also come from the Swedish government.

In summary, our approach demonstrates exceptional performance both in virtual simulation environments and real-world settings, further substantiating the rationality of our assumption regarding the ξ-dependent system in our theorem. In subsection 3.3, we estimated the value of ξ within two scenarios, namely CACC and Real Power-Net 141-bus.

[4] https://www.populationpyramid.net/united-states-of-america/2018/
[5] https://www.lhsfna.org/how-effective-are-masks-and-other-facial-coverings-at-stopping-coronavirus/
[6] https://covid19.who.int/region/euro/country/se

3 Supplemental Result

In this section, we will supplement some ablation experiments, as well as additional findings and results.

3.1 Hyperparameters and model training details

The key hyperparameters and reduction in communication costs for our method are summarized in Table 3. Especially compared to the centralized methods, the communication costs of our method in different scenarios is only about 5%-35% of that in centralized methods.

Supplementary Table 3. Hyperparameters for our algorithm.

	Catch up	Slow down	Eight	Ring	ATSC Grid	Monaco	Power-Grid	Real Power	Pandemic
lr of V_i	3e-4	3e-4	5e-5	5e-4	5e-4	1e-4	5e-4	5e-4	5e-4
lr of π	3e-4	3e-4	5e-5	5e-4	5e-4	1e-4	5e-4	5e-4	2e-4
lr of p_i	3e-4	3e-4	5e-4	5e-4	2e-4	1e-4	5e-4	5e-5	5e-4
π_i Net.	[64,64]	[64,64]	[64,64]	[64,64]	[128,128]	[256,256]	[256,256]	[512,512]	[64,64]
V_i Net.	[64,64]	[64,64]	[64,64]	[64,64]	[128,128]	[256,256]	[256,256]	[512,512]	[64,64]
p_i Net.	[16,16]	[16,16]	[16,16]	[16,16]	[64,64]	[32,32]	[64,64]	[128,128]	[64,64]
κ	1	1	2	2	1	1	1	1	1
Rollout	25	25	25	25	25	25	25	15	25
Cost ($\approx$)	-87%	-87%	-65%	-78%	-83%	-85%	-85%	-95%	-90%

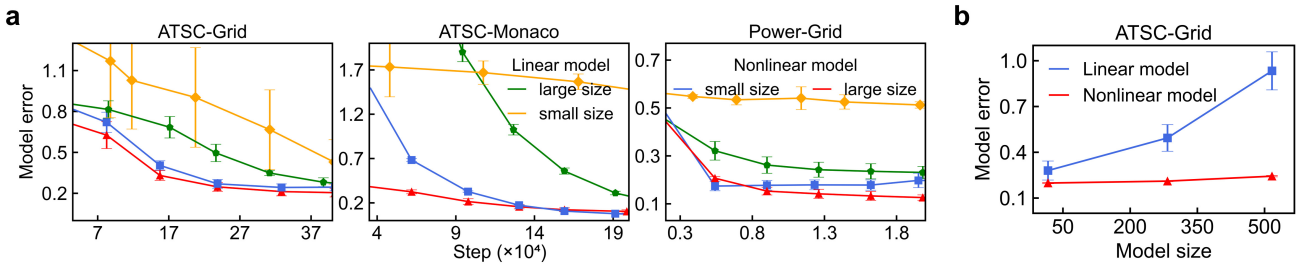
Next, we provide some details of the model learning methods in our method. The universal model error corresponds to the term $D(p\|\hat{p})$ in the inequality (6) in our manuscript. The model error $D(p\|\hat{p})$ can be divided into two parts: dependency bias $D(p\|\bar{p})$ and independence-approximation error $D(\bar{p}\|\hat{p})$. When our model accurately fits the true MDP, we have $D(\bar{p}\|\hat{p}) \approx 0$, and $\sup D(p\|\hat{p}) \approx \sup D(p\|\bar{p}) \leq \xi$. Therefore, $\varepsilon_{m_i} \leq D(p\|\hat{p}) \leq \xi$. We have controlled the model error ε_{m_i} (TV distance between the distributions) under ξ bound through model learning to adapt to our analysis of the monotonic improvement theorem of policy. To obtain a more accurate fitting model, we optimized it from three perspectives: model architecture design, training methods, and training data.

- **Structure of the model** We have undertaken numerous design considerations based on practical engineering experience. Firstly, we meticulously designed the model architecture. **Figure 9** in the main text illustrates the overall structure of our method, wherein the red-bordered box represents the model employed for fitting real-world environments. Concerning the design of the model framework, we distinctively handle state and action information. They are individually subjected to feature extraction using embedding layers, followed by amalgamation based on the topological relationships of the graph network. On one hand, this graph network architecture inherently approximates networked Markov Decision Processes (MDPs). On the other hand, the separate extraction and aggregation of features facilitate rapid learning of environmental dynamical traits by the model. This enables us to maintain a relatively compact network size with reduced model error during training, thereby expediting monotonic policy improvement and minimizing extraneous computational overhead beyond policy learning.

- **Training method and data** We employ a supervised learning approach involving multi-step model extrapolation, iteratively executed in multiple batches. Throughout the model learning process, virtual data generated by the model and authentic environmental data are continually blended in accordance with a predetermined ratio. During the initial stages of training, when the model's accuracy is still developing, the model predominantly relies on authentic environmental data for supervised learning, effectively performing a warm start. As the model attains a certain level of accuracy, we progressively elevate the proportion of model-generated data within the overall training dataset. This ensures that the model effectively fits both mixed real and virtual data. Additionally, guided by disparities between real and virtual data within the amalgamated dataset, a self-supervised mechanism is employed to further adjust model parameters during training iterations. This iterative process facilitates the enhancement of model accuracy and maintains a degree of generalization. During the continuous training of the model, our strategy has also been continuously trained and improved, and our model fitting and strategy improvement are alternating.

3.2 Selection of linear and nonlinear model approximators

We comprehensively compare the performance of linear models and nonlinear neural network models in fitting simple to complex systems, and conclude that linear models are suitable for fitting and analyzing simple linear systems due to their



Supplementary Figure 1. The comparative analysis involves linear models and nonlinear neural networks with varying parameter sizes across distinct environments. a, The model errors in the learning process of models in different scenarios indicate that the fitting ability of linear models is weaker than that of nonlinear neural network models. The data are presented as mean values +/− SD. The center line represents the mean values and the error bar represents the standard deviation ($n = 5$). **b,** When the model parameters are reduced, linear models exhibit a greater reduction in fitting ability compared to nonlinear neural networks. The data are presented as mean values +/− SD. The center line represents the mean values and the error bar represents the standard deviation ($n = 5$).

simplicity and good theoretical characteristics, while nonlinear neural network models are more suitable for model learning in complex large-scale systems, Both in terms of fitting efficiency and fitting accuracy, it has better performance.

To further validate the complexity of the environment, we compared the effectiveness of using linear model $x_{k+1} = Ax_k + Bu_k + C$ to fit the model. For fairness, we used a learnable matrix A, B and C of the same size as the nonlinear neural network model for fitting. From Figure 1(a)-(c), We found that in simple environments and tasks, the final performance of linear model fitting can also meet the control performance requirements, but the fitting efficiency is low and the variance is high, especially when the scene becomes complex. The fitting defects caused by linearity make it difficult to quickly reduce the model fitting error to the ideal state. Compared to linear models in Figure 1(d), the fitting ability of nonlinear neural networks is relatively less sensitive to model size. For better visualization, the model size in Figure 1(d) is defined as $\ln(1/m)$, m represents the number of parameters. Model based reinforcement learning is a dynamic interaction process between policy learning and model learning, and the lower fitting efficiency further affects the learning of strategies, Reduced final control performance. More importantly, When the size of the linear matrix and neural network models decreases simultaneously, the ability of the linear model to fit the model is significantly reduced compared to nonlinear neural networks.

Therefore, from the perspective of reducing model errors as soon as possible, in order to improve the monotonic improvement ability of the policy and reduce the lower bound, we recommend using nonlinear neural networks for fitting in complex task scenarios to better reduce model errors for policy training and model prediction.

3.3 Estimation the range of ξ in complex systems

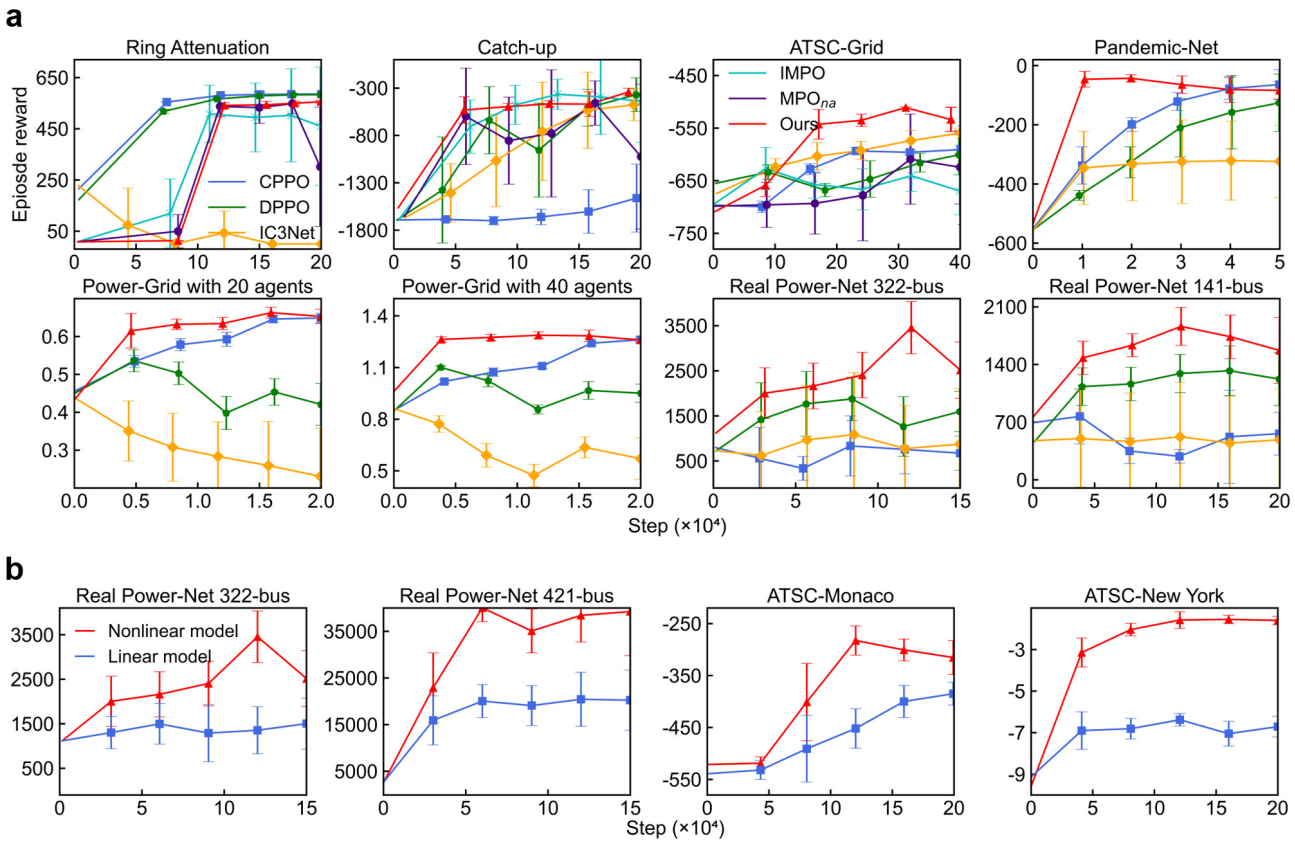
We further validate the rationality of our definitions through experimentation. Specifically, we conducted exhaustive model fitting in both the context of CACC and Real Power-Net, which exhibits a more intricate dynamical dependency. We progressively augmented the parameter count of the fitting models and pursued more comprehensive training to enhance fidelity. Following an extensive series of experiments, we observed that in numerous parameter experiments involving CACC, upon continuous augmentation of model expressiveness and training iterations, further reduction of model error became unattainable. We denote $\widehat{p}(s'|s,a) = \prod_{i=1}^{n} \widehat{p}_i(s'_i|s_{N_i^\kappa}, a_i)$. By using $\widehat{p}(s'|s,a)$ to approximate the true dynamics. At this juncture, we inferred that an optimal local decomposition $\hat{p}^*(s'|s,a)$ for the overall decomposable dynamic transition $\bar{p}$ of the CACC system had been attained:

$$\hat{p}^*(s'|s,a) = \prod_{i=1}^{n} \hat{p}_i^*(s'_i|s_{N_i^\kappa}, a_i). \tag{4}$$

Corresponding to the condition where $D(\bar{p}\|\hat{p}^*) \approx 0$ and $D(p\|\hat{p}^*) > 0$ in the following inequality:

$$D(p\|\hat{p}^*) \leq D(p\|\bar{p}) + D(\bar{p}\|\hat{p}^*). \tag{5}$$

In this inequality, the universal model error $D(p\|\hat{p}^*)$ can be divided into two parts: dependency bias $D(p\|\bar{p})$ and independence-approximation error $D(\bar{p}\|\hat{p}^*)$. According to our **Definition 2** of the ξ-dependent system in the manuscript, when model becomes very accurate, meaning $D(\bar{p}\|\hat{p}^*) \approx 0$, $\sup D(p\|\hat{p}^*) \approx \sup D(p\|\bar{p}) = \xi$. Figure 6 reports that in the CACC and Real



Supplementary Figure 2. The training process in other tasks and the differences in policy learning under nonlinear and linear fitters. a, Training reward increment during the training process under different scenarios and different algorithms. Solid lines represent the mean rewards, and error bars correspond to the standard deviation. The data are presented as mean values $+/-$ SD. The center line represents the mean values and the error bar represents the standard deviation ($n = 5$). **b,** Training reward increment during the training process under linear models and nonlinear models. The data are presented as mean values $+/-$ SD. The center line represents the mean values and the error bar represents the standard deviation ($n = 5$).

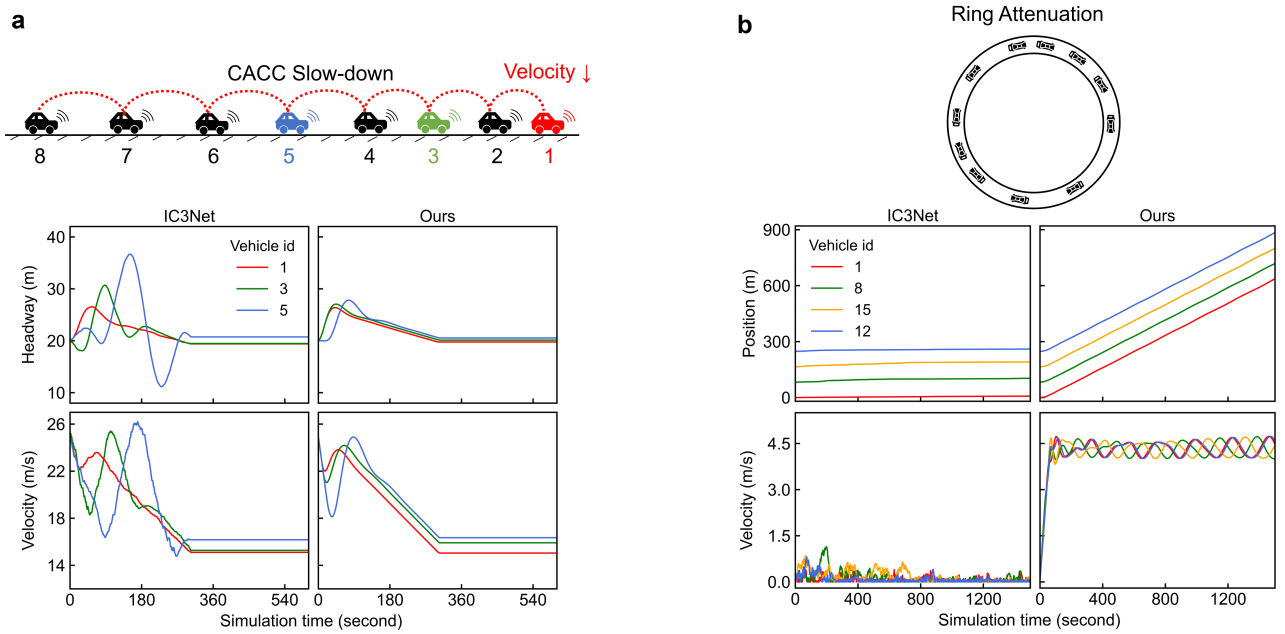
Power-Net systems, with the improvement of parameter quantity and fitting ability, so the model error $D(p\|\hat{p}^*)$ stabilized at around 0.2 and 0.8, respectively. Then we have

$$\sup D(p\|\hat{p}^*) \approx \sup D(p\|\bar{p}) = \xi \geq 0.2 \text{ or } 0.8 > 0 \tag{6}$$

Consequently, we substantiated that $\xi > 0$ in some complex systems, although ξ may be a very small value (this attribute underscores the reliability of our preliminary assumption regarding the independent decomposition of system dynamics in practical systems) in the CACC systems. Analogously, within the more intricate Real Power-Net context, we similarly encountered this phenomenon, manifesting $\xi > 0$. Furthermore, in comparison to the CACC system, the Real Power-Net system exhibits heightened complexity, with inter-agent coupling relationships being more prevalent. As a result, the value of ξ tends to be larger than its value in CACC. Subsequently, in our ensuing theoretical analyses and algorithmic design, we extensively address scenarios where $\xi > 0$, thus providing examples to demonstrate the consistency of our algorithm framework with the ξ-dependent attribute of real multi-agent systems.

3.4 More training results on different tasks

Here we show the training process for all tasks not included in the main text, and from the training process, we can conclude from Fig. 2a that our method has the highest sample efficiency. Especially in large-scale real-world systems, our framework demonstrates superior performance, enabling high decision-making capabilities at low communication costs. In these tasks, the communication cost of our approach is only 10% of that of the centralized algorithm, which makes each decentralized strategy a very small decision network and its training and inference speed far exceeds that of the centralized method. To



Supplementary Figure 3. The evaluation results on vehicle control on CACC and Flow. a, Description of velocity and headway of vehicles 1,3,5 in Catch-up. The visualization of this task is adapted from existing work[6]. The headway stabilizes at 20 m which matches the target headway. The velocity stabilizes at 15 m/s, matching the target velocity. **b,** Description of velocity and position of vehicles 1, 8, 15, 22 in Ring Attention. The visualization of this task is adapted from existing work[13]. Our method controls the formation of queues with a process of accelerating to the safe and efficient velocity near 5 m/s.

achieve performance similar to our method, the centralized method requires that the parameter size of each decision network be at least 10 times that of our method, and the computational efficiency is worrying. In addition, for complex system dynamics in the real world, nonlinear models are not sufficient, which can lead to the failure of policy learning (Fig. 2b).

3.5 More execution results in Connected vehicle control

Figure 3a-b depicts the execution performance of the trained policies based on our method in Flow and CACC. Our analysis suggests that, in comparison to IC3Net, our method effectively governs vehicles to adhere to traffic regulations and uphold safety standards, concurrently enhancing the overall traffic flow efficiency. Specifically in CACC, our method ensures stable control of both velocity and headway of vehicles within the same queue interval, consistently maintaining them around the target value. This leads us to conclude that our algorithm not only improves the efficiency of the overall traffic flow but also ensures safety standards are met.

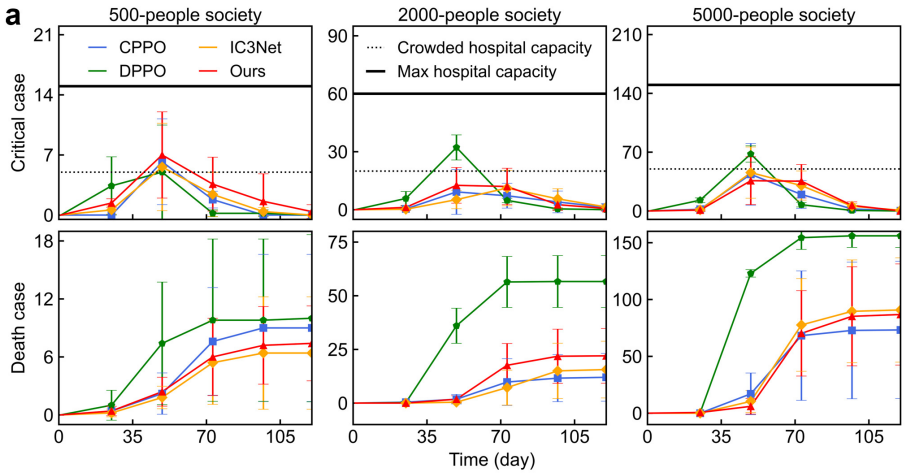
3.6 More execution results in Power-Grid

We further provided the performance of algorithms Only Primary Control, FPrint, CommNet, DIAL and ConseNet in Power-Grid with 20 agents. We also compared the traditional control algorithm MPC. We referred to the evaluation of these methods in previous studies[22]. MPC performance well but it relies heavily on accurate models, which is difficult to achieve in a large number of complex real-world scenarios, and the computational cost of MPC is very high (inference time is about 19 times that of deep reinforcement methods[22]) because it solves online non-linear optimization problems at each time step, requiring a large amount of computation.

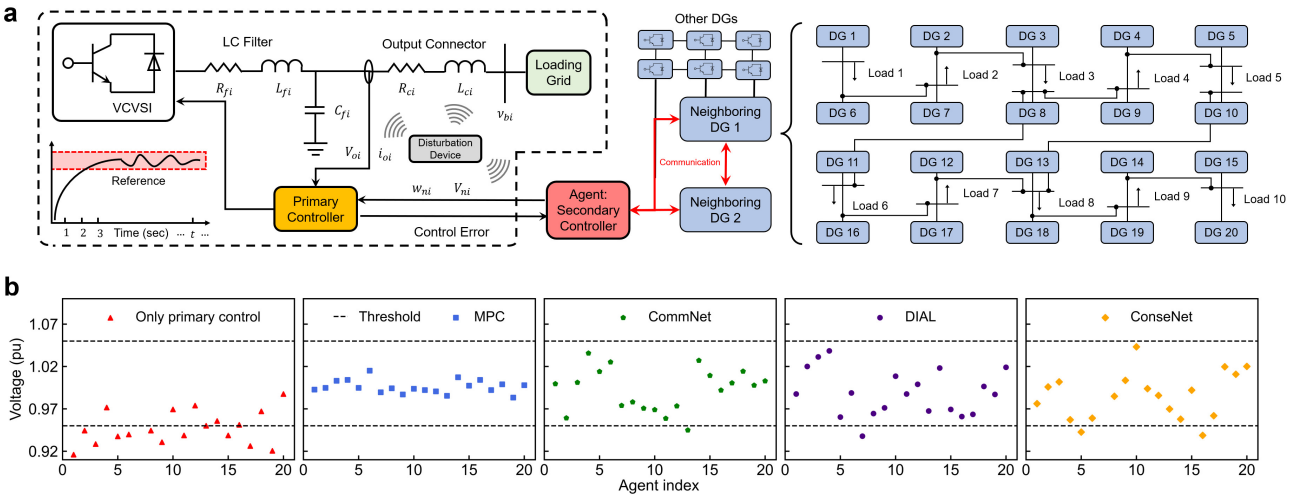
When only Primary Control exists, there is evidence that it is difficult to control the voltage values of all DGs in the system within the reference range, so secondary control is required. We test the performance of different baselines in terms of execution performance is given. As shown in Figure 5, other baselines are difficult to control all DGs within the voltage reference range.

3.7 More execution results in Pandemic networks

In the large-scale spread of the epidemic in society, real-time local decision-making is very important. As the infection situation gradually worsens, the decisions of different decision-making institutions (education departments, medical systems, social service systems, etc.) usually rely on the local information possessed by the current institutions. We observe the overall

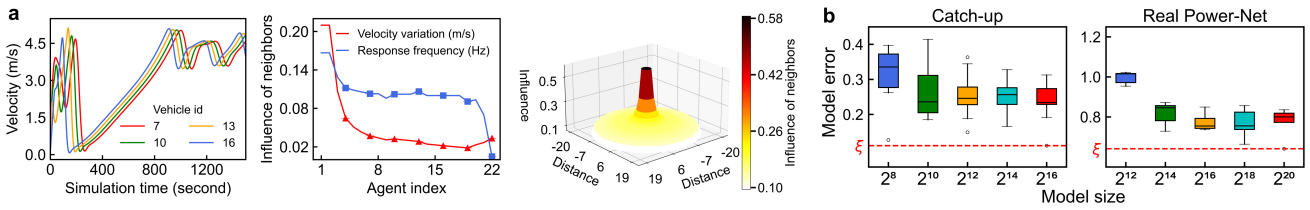


Supplementary Figure 4. Execution performance of the final decision models in pandemic network a, The number of critical cases and death cases when using strategies provided by different algorithms in societies with different population sizes. The data are presented as mean values +/− SD. The center line represents the mean values and the error bar represents the standard deviation ($n = 5$).



Supplementary Figure 5. Description of the secondary control system of the power-grid and more execution results. a, Schematic diagram of the decentralized control of power-grid. This system architecture is adapted from existing work[22]. **b,** Decision performance of only primary control, traditional control algorithm MPC and other reinforcement learning algorithms.

infection of different infectious disease prevention and control strategies in a society with a growing population in three months. Figure 4a reports the relationship between severe cases and the maximum capacity of hospitals under different policies, If the number of severe cases exceeds the maximum capacity of the hospital, it indicates that the current virus transmission is very serious. Our algorithm can effectively control severe cases below the maximum capacity of the hospital and alleviate the pressure on medical staff. Our method has also maintained regional economic development as much as possible. These indicators verify that our method can not only alleviate the harm caused by virus transmission to human health in real pandemic systems but also balance the economic development of large societies.



Supplementary Figure 6. Analysis of system attenuation characteristics and verification of ξ-dependent system assumption a, We show the influence of the following vehicles when the leading vehicle suddenly accelerates or decelerates. We can conclude that the influence will gradually weaken as the network topology distance increases. we also show the relationship between the distance and influence between agents under different κ in more general networked systems. The overall trend is that the influence between agents gradually attenuates from the central agent to the surroundings. **b,** The relationship between the limit error of model fitting and the model's fitting capacity (such as the number of parameters). The abscissa represents the number of hidden layers and neurons in the principal structural components of the neural network employed for model fitting, ranging from 16×16 to 1024×1024 The data ($n = 5$) are presented as median values (the central line of each box) along with the 25th and 75th percentiles (the bottom and top edges of each box), minima and maxima (the whiskers attached to each box), as well as outliers (outside the box and whiskers). The ordinate denotes the ultimate discrepancy between the fitted model and the actual environment under the present parameter configuration.

4 Details of Proofs

4.1 Remarks on Proofs of Lemmas and Theorems

To bound the returns, we use TV distance to assess the distance between s_i, a_i at every timestep, based on the fact that $|\mathbb{E}_{s_i,a_i\sim p_1} r(s_i,a_i) - \mathbb{E}_{s_i,a_i\sim p_2} r(s_i,a_i)| \leq r_{\max} \cdot D_{TV}(p_1 \| p_2)$. To achieve this, we argue that in an INS, s_i^T only depends on $s_{N_i^1}^{T-1}$, which is the 1-hop neighbors' states at previous timestep. Then, inductively, s_i^T is only dependent on $s_{N_i^1}^{T-1}, s_{N_i^2}^{T-2}, ..., s_{N_i^k}^{T-k}$, and ultimately, $s_{N_i^T}^0$. Then we consider a chain of states $x^t = s_{N_i^{N-t}}, t = 0, 1, ..., N$ in Lemma 4 to analyze how s_i^t depends on model errors and policy divergences in a step-by-step manner. Insipred by[15], by summing the divergence at each timestep $t = 0, 1, ...$, we can generate a overall discounted returns bound of branched rollout in Lemma 5. Lemma 6 is a simple corollary of Lemma 5, which handles normal rollout scenario.

If the returns are bounded, we can show that if every optimization step increase the return by C, the real return would also almost monotonically increase. Specifically, if $\hat{\eta}[\pi^{k+1}] \geq \hat{\eta}[\pi^k] + C$, combined with $\eta[\pi^{k+1}] \geq \hat{\eta}[\pi^k] - C$, we have $\eta[\pi^{k+1}] \geq \hat{\eta}[\pi^k]$. But note that $\hat{\eta}[\pi^k]$ increases monotonically, and it's a lower bound of $\eta[\pi^{k+1}]$. This means that $\hat{\eta}[\pi^{k+1}]$ would almost monotonically increase.

4.2 Proof of Theorem 1

$\eta^p[\pi_1, ..., \pi_n]$ and $\eta^{\hat{p}}[\pi_1, ..., \pi_n]$ differs in two ways. First, they are estimated with different transition dynamics. Second, the latter sample their states and actions with another policy, namely π_D. To deal with these divergences separately, we use an intermediate return $\eta[\pi_D]$, which is the return of π_D in the environment. Then the difference can be bounded by:

$$|\eta_i^p[\pi_1, ..., \pi_n] - \eta_i^{\hat{p}}[\pi_1, ..., \pi_n]| \leq \underbrace{|\eta_i^p[\pi_1, ..., \pi_n] - \eta_i^p[\pi_D]|}_{L_1} + \underbrace{|\eta_i^p[\pi_D] - \eta_i^{\hat{p}}[\pi_1, ..., \pi_n]|}_{L_2}. \tag{7}$$

To bound L_1, we apply Lemma 6 with $\varepsilon_{m_i} = 0$. Then:

$$L_1 \leq \frac{2r_{\max}}{1-\gamma}\big[\varepsilon_{\pi_i} + \sum_{k=0}^{\infty} \gamma^{k+1} \sum_{j\in N_i^k} \varepsilon_{\pi_j}\big]. \tag{8}$$

In L_2, both the dynamics and policies and different, therefore:

$$L_2 \leq \frac{2r_{\max}}{1-\gamma}\big[\varepsilon_{\pi_i} + \sum_{k=0}^{\infty} \gamma^{k+1} \sum_{j\in N_i^k} (\varepsilon_{m_j} + \varepsilon_{\pi_j})\big]. \tag{9}$$

Putting Equation 8 and 9 yields:

$$|\eta_i^p[\pi_1, ..., \pi_n] - \eta_i^{\hat{p}}[\pi_1, ..., \pi_n]| \leq \frac{2r_{\max}}{1-\gamma}\big[2\varepsilon_{\pi_i} + \sum_{k=0}^{\infty} \gamma^{k+1} \sum_{j\in N_i^k} (\varepsilon_{m_j} + 2\varepsilon_{\pi_j})\big]. \tag{10}$$

And because the binary relation of k-hop neighbor is symmetric, we have:

$$\frac{1}{n}\sum_{i=1}^{n}\sum_{j\in N_i^k} \varepsilon_j = \sum_{j=1}^{n} \frac{|V_j^k|}{n}\varepsilon_j. \tag{11}$$

Then, by averaging Equation 10, we have:

$$\begin{aligned}
|\eta^p[\pi_1, ..., \pi_n] - \eta^{\hat{p}}[\pi_1, ..., \pi_n]| &\leq \frac{1}{n}\sum_{i=1}^{n} |\eta_i^p[\pi_1, ..., \pi_n] - \eta_i^{\hat{p}}[\pi_1, ..., \pi_n]| \\
&\leq \frac{2r_{\max}}{1-\gamma}\big[\frac{1}{n}\sum_{i=1}^{n}\varepsilon_{\pi_i} + \sum_{k=0}^{\infty}\gamma^{k+1}\frac{1}{n}\sum_{i=1}^{n}\sum_{j\in N_i^k}(\varepsilon_{m_j} + 2\varepsilon_{\pi_j})\big] \\
&= \frac{2r_{\max}}{1-\gamma}\big[\frac{1}{n}\sum_{i=1}^{n}\varepsilon_{\pi_i} + \sum_{k=0}^{\infty}\gamma^{k+1}\sum_{i=1}^{n}\frac{|N_i^k|}{n}(\varepsilon_{m_i} + 2\varepsilon_{\pi_i})\big] \\
&= \frac{2r_{\max}}{1-\gamma}\sum_{i=1}^{n}\big[\frac{\varepsilon_{\pi_i}}{n} + (\varepsilon_{m_i} + 2\varepsilon_{\pi_i})\cdot\sum_{k=0}^{\infty}\gamma^{k+1}\frac{|N_i^k|}{n}\big]
\end{aligned} \tag{12}$$

4.3 Proof of Theorem 2

To bound $|\eta^p[\pi_1,...,\pi_n] - \eta^{branch}[\pi_1,...,\pi_n]|$, we need to analyze how do they differ from each other. η^p denote the real returns of these policies in the environment. η^{branch} is the returns estimated in the branched rollout scheme. To explicitly illustrate this point, we describe their difference in Table 4:

branch point	before		after	
	dynamics	policies	dynamics	policies
η^p	p	π	p	π
η^{branch}	p	π_D	$\hat{p}$	π

Supplementary Table 4. The difference between η^p and η^{branch}

By Lemma 5, we have:

$$|\eta_i^p[\pi_1,...,\pi_n] - \eta_i^{branch}[\pi_1,...,\pi_n]| \leq \frac{2r_{\max}}{1-\gamma}\Big[\sum_{k=0}^{T-1}\gamma^{k+1}\sum_{j\in N_i^k}\varepsilon_{m_j} + \sum_{k=T}^{\infty}\gamma^{k+1}\sum_{j\in N_i^k}\varepsilon_{\pi_j}\Big] \tag{13}$$

And because the binary relation of k-hop neighbor is symmetric, we have:

$$\frac{1}{n}\sum_{i=1}^{n}\sum_{j\in N_i^k}\varepsilon_j = \sum_{j=1}^{n}\frac{|V_j^k|}{n}\varepsilon_j. \tag{14}$$

Then, by averaging Equation 13, we have:

$$\begin{aligned}
|\eta^p[\pi_1,...,\pi_n] - \eta^{branch}[\pi_1,...,\pi_n]| &\leq \frac{1}{n}\sum_{i=1}^{n}|\eta_i^p[\pi_1,...,\pi_n] - \eta_i^{branch}[\pi_1,...,\pi_n]| \\
&\leq \frac{2r_{\max}}{1-\gamma}\Big[\sum_{k=0}^{T-1}\gamma^{k+1}\cdot\frac{1}{n}\sum_{i=1}^{n}\sum_{j\in N_i^k}\varepsilon_{m_j} + \sum_{k=T}^{\infty}\gamma^{k+1}\cdot\frac{1}{n}\sum_{i=1}^{n}\sum_{j\in N_i^k}\varepsilon_{\pi_j}\Big] \\
&= \frac{2r_{\max}}{1-\gamma}\Big[\sum_{k=0}^{T-1}\gamma^{k+1}\cdot\sum_{i=1}^{n}\frac{|N_i^k|}{n}\varepsilon_{m_i} + \sum_{k=T}^{\infty}\gamma^{k+1}\cdot\sum_{i=1}^{n}\frac{|N_i^k|}{n}\varepsilon_{\pi_i}\Big] \\
&= \frac{2r_{\max}}{1-\gamma}\sum_{i=1}^{n}\Big[\varepsilon_{m_i}\cdot\Big(\sum_{k=0}^{T-1}\gamma^{k+1}\frac{|N_i^k|}{n}\Big) + \varepsilon_{\pi_i}\cdot\Big(\sum_{k=T}^{\infty}\gamma^{k+1}\frac{|N_i^k|}{n}\Big)\Big]
\end{aligned} \tag{15}$$

4.4 Proof of Corollary 1

Although $p(s'|s,a)$ might not satisfy the criteria of an INS, we can construct another transition dynamic $\hat{p} = \prod_{i=1}^{n} p_i(s_i|s_{N_i},a_i)$, which is the product of marginal transitions, thus being an INS. Recall that by definition, if p is a transition dynamic of a ξ-dependent system, $\sup_{s,a} D_{TV}(p(s'|s,a)\|\hat{p}(s'|s,a)) \leq \xi$. Then we divide the difference into two parts:

$$\eta^p[\pi_1,...,\pi_n] - \eta^{branch}[\pi_1,...,\pi_n] = \underbrace{\eta^p[\pi_1,...,\pi_n] - \eta^{\hat{p}}[\pi_1,...,\pi_n]}_{L_1} + \underbrace{\eta^{\hat{p}}[\pi_1,...,\pi_n] - \eta^{branch}[\pi_1,...,\pi_n]}_{L_2} \tag{16}$$

The first part is the difference of policy returns, estimated in two environments. Lemma 2 gives that:

$$D_{TV}\left(p^t(s,a)\|\hat{p}^t(s,a)\right) \leq D_{TV}\left(p^0(s,a)\|\hat{p}^0(s,a)\right) + \sum_{i=1}^{t} D_{TV}\left(p^i(s^i,a^i|s^{i-1},a^{i-1})\|\hat{p}^i(s^i,a^i|s^{i-1},a^{i-1})\right). \tag{17}$$

Note that the first part has no policy divergences, which follows from Lemma 1, thus

$$\begin{aligned}
D_{TV}\left(p^i(s^i,a^i|s^{i-1},a^{i-1})\|\hat{p}^i(s^i,a^i|s^{i-1},a^{i-1})\right) &\leq D_{TV}\left(p^i(s^i|s^{i-1},a^{i-1})\|\hat{p}^i(s^i|s^{i-1},a^{i-1})\right) + D_{TV}\left(\pi(a^i|s^i)\|\pi(a^i|s^i)\right) \\
&\leq \xi + 0 = \xi.
\end{aligned} \tag{18}$$

Plug (18) into (17) yields

$$D_{TV}\left(p^t(s,a)\|\hat{p}^t(s,a)\right) \leq D_{TV}\left(p^0(s,a)\|\hat{p}^0(s,a)\right) + \sum_{i=1}^{t}\xi = t\xi. \tag{19}$$

Then, we have:

$$\begin{aligned} L_1 = \eta^p[\pi_1,...,\pi_n] - \eta^{\hat{p}}[\pi_1,...,\pi_n] &\leq 2r_{\max}\sum_{t=0}^{\infty}\gamma^t D_{TV}\left(p^t(s,a)\|\hat{p}^t(s,a)\right) \\ &\leq 2r_{\max}\sum_{t=0}^{\infty}\gamma^t t\xi \\ &= \frac{2r_{\max}}{1-\gamma}\cdot\frac{\gamma}{1-\gamma}\xi. \end{aligned} \tag{20}$$

As for L_2, because $\hat{p}$ is an INS, we can directly apply Theorem 2:

$$L_2 \leq \frac{2r_{\max}}{1-\gamma}\sum_{i=1}^{n}\left[\varepsilon_{m_i}\cdot\left(\sum_{k=0}^{T-1}\gamma^{k+1}\frac{|N_i^k|}{n}\right)+\varepsilon_{\pi_i}\cdot\left(\sum_{k=T}^{\infty}\gamma^{k+1}\frac{|N_i^k|}{n}\right)\right] \tag{21}$$

Summation of Equation 20 and Equation 21 completes the proof.

4.5 Proof of Theorem 3

In[14] (Lemma 3), it was proven that if $s_{N_i^\kappa}$ and $a_{N_i^\kappa}$ are fixed, then no matter how distant states and actions changes, Q-function will not change significantly:

$$|Q_i(s_{N_i^\kappa},a_{N_i^\kappa},s_{N_{-i}^\kappa},a_{N_{-i}^\kappa}) - Q_i(s_{N_i^\kappa},a_{N_i^\kappa},s'_{N_{-i}^\kappa},a'_{N_{-i}^\kappa})| \leq \frac{r_{\max}}{1-\gamma}\gamma^\kappa.$$

As the value function is the expectation of the Q-function

$$\begin{aligned} V_i(s) &= \mathbb{E}_{a\sim\pi}Q_i(s,a) \\ V_i(s_{N_i^\kappa}) &= \mathbb{E}_{a\sim\pi}Q_i(s_{N_i^\kappa},a_{N_i^\kappa}), \end{aligned}$$

we have,

$$\begin{aligned} |V_i(s) - V_i(s_{N_i^\kappa})| &= |\mathbb{E}_{a\sim\pi}Q_i(s,a) - \mathbb{E}_{a\sim\pi}Q_i(s_{N_i^\kappa},a_{N_i^\kappa})| \\ &\leq \mathbb{E}_{a\sim\pi}|Q_i(s,a) - Q_i(s_{N_i^\kappa},a_{N_i^\kappa})| \\ &\leq \frac{r_{\max}}{1-\gamma}\gamma^\kappa, \end{aligned}$$

which concludes the proof. This theoretical result is based on previous research[23].

4.6 Proof of Theorem 4

The difference of the gradients is written as

$$\begin{aligned} g_i - \tilde{g}_i =& \mathbb{E}(\hat{A}-\tilde{A})\nabla_{\theta_i}\log\pi_i(a_i|s_{N_i}) \\ =& \frac{1}{n}\mathbb{E}[\sum_{j\notin N_i^\kappa}\hat{A}_j]\nabla_{\theta_i}\log\pi_i(a_i|s_{N_i}) + \frac{1}{n}\mathbb{E}[\sum_{j\in N_i^\kappa}(\hat{A}_j-\tilde{A}_j)]\nabla_{\theta_i}\log\pi_i(a_i|s_{N_i}) \\ =& \frac{1}{n}\mathbb{E}[\sum_{j\notin N_i^\kappa}(r_j+\gamma V_j(s')-V_j(s))]\nabla_{\theta_i}\log\pi_i(a_i|s_{N_i}) + \frac{1}{n}\mathbb{E}\sum_{j\in N_i^\kappa}\big[(r_j+\gamma V_j(s')-V_j(s)) \\ & -(r_j+\gamma V_j(s'_{N_i^\kappa})-V_j(s_{N_i^\kappa}))\big]\nabla_{\theta_i}\log\pi_i(a_i|s_{N_i}) \\ =& L_1+L_2. \end{aligned} \tag{22}$$

Note that for any function $b(s)$, $\mathbb{E}[b(s)\nabla_{\theta_i}\log\pi^{\theta_i}(a_i|s_{N_i})]=0$. Therefore, L_2 in Equation (22) becomes:

$$\begin{aligned} |L_2| \leq& \frac{1}{n}\mathbb{E}\sum_{j\in N_i^\kappa}\gamma|V_j(s')-V_j(s'_{N_i^\kappa})]||\nabla_{\theta_i}\log\pi_i(a_i|s_{N_i})|| \\ \leq& \frac{|N_i^\kappa|}{n}\frac{\gamma^{\kappa+1}}{1-\gamma}r_{\max}g_{\max}. \end{aligned} \tag{23}$$

For L_1, note that $r_j + \gamma V_j(s') - V_j(s) = -V_j(s) + r_j + \sum_{t=1}^{\kappa-2} \mathbb{E}\gamma^t r_j^t + \gamma^{\kappa-1} V_j(s^{\kappa-1})$. And in an independent network system (where Eq.(4) holds), $s_j^t, a_j^t, t = 0, 1, ..., \kappa-2$ is not affected by policy π_i if $j \notin N_i^\kappa$, we have that

$$\begin{aligned} |L_1| \leq & \frac{1}{n} \mathbb{E} \sum_{j \notin N_i^\kappa} |\gamma^{\kappa-1} V_j(s^{\kappa-1})| |\nabla_{\theta_i} \log \pi_i(a_i|s_{N_i})|| \\ \leq & (1 - \frac{|N_i^\kappa|}{n}) \frac{\gamma^{\kappa-1}}{1-\gamma} r_{\max} g_{\max}. \end{aligned} \tag{24}$$

Put Equation (23) and (24) together, we have

$$\begin{aligned} |g_i - \tilde{g}_i| \leq & |L_1| + |L_2| \\ \leq & \frac{|N_i^\kappa|}{n} \frac{\gamma^{\kappa+1}}{1-\gamma} r_{\max} g_{\max} + (1 - \frac{|N_i^\kappa|}{n}) \frac{\gamma^{\kappa-1}}{1-\gamma} r_{\max} g_{\max} \\ = & \frac{\gamma^{\kappa-1}}{1-\gamma} [1 - (1-\gamma^2) \frac{|N_i^\kappa|}{n}] r_{\max} g_{\max}. \end{aligned}$$

4.7 Proof of Lemmas

Below we give Lemmas that are used in the theorems and the their proofs. Note that Lemma 1 was firstly proposed and proved in[15]. We list it here to make the presentation self-contained.

Lemma 1. *(TVD of Joint Distribution[15]) Consider two distributions $p_1(x,y) = p_1(x)p_1(y|x)$ and $p_2(x,y) = p_2(x)p_2(y|x)$. The total variation distance between them can be bounded as:*

$$\begin{aligned} D_{TV}\big(p_1(x,y)\|p_2(x,y)\big) &\le D_{TV}\big(p_1(x)\|p_2(x)\big) + \mathbb{E}_{x\sim p_2}\big[D_{TV}\big(p_1(y|x)\|p_2(y|x)\big)\big] \\ &\le D_{TV}\big(p_1(x)\|p_2(x)\big) + \max_x D_{TV}\big(p_1(y|x)\|p_2(y|x)\big) \end{aligned}$$

Lemma 2. *Suppose there are two chains of distributions $\{x_1^t, t\ge 0\}, \{x_2^t, t\ge 0\}$. At time t, the states of both chains share an identical state space $x_1^t, x_2^t \in \mathscr{X}^t$. Suppose these two chains satisfy a Markov-like property: $p^{t+1}(x^{t+1}|x^t,...,x^1,x^0) = p^{t+1}(x^{t+1}|x^t)$. Then, the TVD of distributions of two chains at time t can be decomposed as:*

$$D_{TV}\big(p_1^T(x^T)\|p_2^T(x^T)\big) \le D_{TV}\big(p_1^0(x^0)\|p_2^0(x^0)\big) + \sum_{t=1}^{T}\mathbb{E}_{s^{t-1}\sim p_2^{t-1}} D_{TV}\big(p_1^t(x^t|x^{t-1})\|p_2^t(x^t|x^{t-1})\big).$$

We prove this lemma by induction. When $T=0$, it's easy to see that this lemma is true.
Assume it is true for $T=k$. We have:

$$\begin{aligned} |p_1^{k+1}(x^{k+1}) - p_2^{k+1}(x^{k+1})| =& |\sum_{x^k}[p_1^{k+1}(x^{k+1}|x^k)p_1^k(x^k) - p_2^{k+1}(x^{k+1}|x^k)p_2^k(x^k)]| \\ \le& \sum_{x^k}|p_1^{k+1}(x^{k+1}|x^k)p_1^k(x^k) - p_2^{k+1}(x^{k+1}|x^k)p_2^k(x^k)| \\ =& \sum_{x^k}|p_1^{k+1}(x^{k+1}|x^k)p_1^k(x^k) - p_1^{k+1}(x^{k+1}|x^k)p_2^k(x^k) + p_1^{k+1}(x^{k+1}|x^k)p_2^k(x^k) - p_2^{k+1}(x^{k+1}|x^k)p_2^k(x^k)| \\ \le& \sum_{x^k}\big[p_2^k(x^k)|p_1^{k+1}(x^{k+1}|x^k) - p_2^{k+1}(x^{k+1}|x^k)| + p_1^{k+1}(x^{k+1}|x^k)|p_1^k(x^k) - p_2^k(x^k)|\big] \\ =& \mathbb{E}_{x^k\sim p_2^k}[|p_1^{k+1}(x^{k+1}|x^k) - p_2^{k+1}(x^{k+1}|x^k)|] + \sum_{x^k} p_1^{k+1}(x^{k+1}|x^k)|p_1^k(x^k) - p_2^k(x^k)| \end{aligned}$$

Then,

$$\begin{aligned} D_{TV}\big(p_1^{k+1}(x^{k+1})\|p_2^{k+1}(x^{k+1})\big) =& \frac{1}{2}\sum_{x^{k+1}}|p_1^{k+1}(x^{k+1}) - p_2^{k+1}(x^{k+1})| \\ \le& \frac{1}{2}\sum_{x^{k+1}}\Big(\mathbb{E}_{x^k\sim p_2^k}[|p_1^{k+1}(x^{k+1}|x^k) - p_2^{k+1}(x^{k+1}|x^k)|] + \sum_{x^k} p_1^{k+1}(x^{k+1}|x^k)|p_1^k(x^k) - p_2^k(x^k)|\Big) \\ =& \mathbb{E}_{s^k\sim p_2^k} D_{TV}\big(p_1^{k+1}(x^{k+1}|x^k)\|p_2^{k+1}(x^{k+1}|x^k)\big) + D_{TV}\big(p_1^k(x^k)\|p_2^k(x^k)\big) \\ \le& D_{TV}\big(p_1^0(x^0)\|p_2^0(x^0)\big) + \sum_{t=1}^{k+1}\mathbb{E}_{s^{t-1}\sim p_2^{t-1}} D_{TV}\big(p_1^t(x^t|x^{t-1})\|p_2^t(x^t|x^{t-1})\big) \end{aligned}$$

Then this theorem holds for all $n\in\mathbb{N}$.

Lemma 3. *Consider two distributions with pdf $p(x)$ and $q(x)$, where $x = (x_1,...,x_n)\in\mathbb{R}^n$. Suppose p and q can be factorized as: $p(x) = \prod_{i=1}^n p_i(x_i), q(x) = \prod_{i=1}^n q_i(x_i)$. Then we have:*

$$D_{TV}[p(x)\|q(x)] \le \sum_{i=1}^n D_{TV}[p_i(x_i)\|q_i(x_i)].$$

Also, if the distance is measured by KL-divergence, we have:

$$D_{KL}[p(x)\|q(x)] = \sum_{i=1}^n D_{KL}[p_i(x_i)\|q_i(x_i)].$$

We prove this result for discrete distributions, yet by replacing sum with integration, this result stays true in continuous case.

$$\begin{aligned}
D_{TV}[p(x)\|q(x)] = \frac{1}{2}\sum_x |p(x)-q(x)| &= \frac{1}{2}\sum_{x_1,\dots,x_n} |\prod_{k=1}^n p_k(x_k) - \prod_{k=1}^n q_k(x_k)| \\
&= \frac{1}{2}\sum_{x_1,\dots,x_n} \Big|\sum_{i=1}^n \Big[\prod_{k=1}^{i-1} p_k(x_k) \prod_{k=i+1}^n q_k(x_k)(p_i(x_i)-q_i(x_i))\Big]\Big| \\
&\le \frac{1}{2}\sum_{x_1,\dots,x_n}\sum_{i=1}^n \Big[\prod_{k=1}^{i-1} p_k(x_k) \prod_{k=i+1}^n q_k(x_k)|p_i(x_i)-q_i(x_i)|\Big] \\
&= \frac{1}{2}\sum_{i=1}^n \sum_{x_i} |p_i(x_i)-q_i(x_i)| \\
&= \sum_{i=1}^n D_{TV}[p_i(x_i)\|q_i(x_i)].
\end{aligned}$$

In KL-divergence case, we have:

$$\begin{aligned}
D_{KL}[p(x)\|q(x)] &= \sum_x p(x)\log\frac{p(x)}{q(x)} \\
&= \sum_{x_1,\dots,x_n}\Big[p(x_1,\dots,x_n)\sum_{i=1}^n \log\frac{p_i(x_i)}{q_i(x_i)}\Big] \\
&= \sum_{i=1}^n \sum_{x_1,\dots,x_n}\Big[\prod_{k=1}^n p_k(x_k)\log\frac{p_i(x_i)}{q_i(x_i)}\Big] \\
&= \sum_{i=1}^n \sum_{x_i} p_i(x_i)\log\frac{p_i(x_i)}{q_i(x_i)} \\
&= \sum_{i=1}^n D_{KL}(p_i(x_i)\|q_i(x_i)).
\end{aligned}$$

Lemma 4. *(N-step distribution distance) Suppose the expected TVD between two dynamics transitions is bounded as* $\varepsilon_{m_i} = \max_{s_{N_i},a_i} D_{TV}\left[p(s_i'|s_{N_i},a_i)\|\hat{p}(s_i'|s_{N_i},a_i)\right]$ *and* $\varepsilon_{\pi_i} = \max_s D_{TV}\left[\pi_i(a_i|s_{N_i})\|\hat{\pi}_i(a_i|s_{N_i})\right]$*. Then the N-step distribution distance is bounded as:*

$$D_{TV}\left[p_i^N(s_i)\|\hat{p}_i^N(s_i)\right] \le \sum_{t=0}^{N-1}\sum_{j\in N_i^t}(\varepsilon_{\pi_j}+\varepsilon_{m_j}).$$

Thus,

$$D_{TV}\left[p_i^N(s_i,a_i)\|\hat{p}_i^N(s_i,a_i)\right] \le \varepsilon_{\pi_i} + \sum_{t=0}^{N-1}\sum_{j\in N_i^t}(\varepsilon_{\pi_j}+\varepsilon_{m_j}).$$

Consider a chain of regional state $x^t = s_{N_i^{N-t}}$. Two chains of distributions $p^t(x^t), \hat{p}(x^t)$ denote the distributions of x^t under the environment and our model, respectively. First, because of the property of an INS, these two chains' transition dynamics can be decomposed as:

$$\begin{aligned}
p(x^t|x^{t-1}) &= p(s'_{N_i^{N-t}}|s_{N_i^{N-t+1}}) \\
&= \prod_{j\in N_i^{N-t}} p_j(s_j'|s_{V_j}).
\end{aligned}$$

And because $p_j(s_j'|s_{V_j}) = \sum_{a_j} p_j(s_j'|s_{V_j},a_j)\pi_j(a_j|s_{V_j})$, by the property of TVD, we know that:

$$D_{TV}\left[p_j(s_j'|s_{V_j})\|\hat{p}_j(s_j'|s_{V_j})\right] \le \varepsilon_{\pi_j}+\varepsilon_{m_j}.$$

With Lemma 3, we have:

$$D_{TV}\left[p(x^t|x^{t-1})\|\hat{p}(x^t|x^{t-1})\right] \leq \sum_{j\in N_i^{N-t}} (\varepsilon_{\pi_j} + \varepsilon_{m_j}). \tag{25}$$

Then we are ready to decompose $D_{TV}\left[p_i^N(s_i)\|\hat{p}_i^N(s_i)\right]$.
By definition, $x^N = s_{N_i^{N-N}} = s_i$, then

$$D_{TV}\left[p_i^N(s_i)\|\hat{p}_i^N(s_i)\right] = D_{TV}\left[p_i^N(x^N)\|\hat{p}_i^N(x^N)\right].$$

Then the two chains of distributions $p(x^0),...,p(x^N)$ and $\hat{p}(x^0),...,\hat{p}(x^N)$ are Markov chains, because by definition,

$$\begin{aligned} p(x^{t+1}|s) &= p(s_{N_i^{N-t-1}}|s) \\ &= p(s_{N_i^{N-t-1}}|s_{N_i^{N-t}}) \\ &= p(x^{t+1}|x^t), \end{aligned}$$

which means x^{t+1} is solely dependent on x^t.

Applying Lemma 2 on these chains, we have

$$\begin{aligned} D_{TV}\left[p_i^N(s_i)\|\hat{p}_i^N(s_i)\right] &= D_{TV}\left[p_i^N(x^N)\|\hat{p}_i^N(x^N)\right] \\ &\leq D_{TV}(p_1^0(x^0)\|p_2^0(x^0)) + \sum_{t=1}^{N} \mathbb{E}_{x^{t-1}\sim p_2^{t-1}} D_{TV}\left(p_1^t(x^t|x^{t-1})\|p_2^t(x^t|x^{t-1})\right). \end{aligned}$$

By (25), we have

$$\begin{aligned} D_{TV}\left[p_i^N(s_i)\|\hat{p}_i^N(s_i)\right] &\leq D_{TV}(p_1^0(x^0)\|p_2^0(x^0)) + \sum_{t=1}^{N} \mathbb{E}_{x^{t-1}\sim p_2^{t-1}} D_{TV}\left(p_1^t(x^t|x^{t-1})\|p_2^t(x^t|x^{t-1})\right) \\ &\leq 0 + \sum_{t=1}^{N} \sum_{j\in N_i^{N-t}} (\varepsilon_{\pi_j} + \varepsilon_{m_j}) \\ &= \sum_{t=0}^{N-1} \sum_{j\in N_i^t} (\varepsilon_{\pi_j} + \varepsilon_{m_j}). \end{aligned}$$

which completes the proof of the first part. And by Lemma 1, the second part holds true.

Lemma 5. *(Returns bound measured in branched rollout) Consider two MDPs $p(s)$ and $\hat{p}(s)$. Suppose they both adopt T-branched rollout scheme. Before the branch, suppose the dynamics distributions are bounded as:*

$$\max_{s_{N_i},a_i} D_{TV}\left(p_i^{pre}(s_i'|s_{N_i},a_i)\|\hat{p}_i^{pre}(s_i'|s_{N_i},a_i)\right) = \varepsilon_{m_i}^{pre}$$

and policy divergences are bounded as:

$$\max_{s_{N_i}} D_{TV}\left(\pi_i^{pre}(a_i|s_{N_i})\|\hat{\pi}_i^{pre}(a_i|s_{N_i})\right) = \varepsilon_{\pi_i}^{pre}$$

After the branch, $\varepsilon_{m_i}^{post}$ and $\varepsilon_{\pi_i}^{post}$ are defined similarly. Then the (local) T-step returns are bounded as:

$$|\eta_i - \hat{\eta}_i| \leq \frac{2r_{\max}}{1-\gamma}\left[\varepsilon_{\pi_i}^{post} + \sum_{k=0}^{T-1} \gamma^{k+1} \sum_{j\in N_i^k} (\varepsilon_{m_j}^{post} + \varepsilon_{\pi_j}^{post}) + \sum_{k=T}^{\infty} \gamma^{k+1} \sum_{j\in N_i^k} (\varepsilon_{m_j}^{pre} + \varepsilon_{\pi_j}^{pre})\right]$$

We prove this by estimating the state-action distribution divergence at each timestep. For notation simplicity, we denote $\varepsilon_{N_i^k}^{pre} = \varepsilon_{N_i^k}^{pre} = \sum_{j\in N_i^k}(\varepsilon_{m_j}^{pre} + \varepsilon_{\pi_j}^{pre})$, and $\varepsilon_{N_i^k}^{post}$ analogously. If we divide the chains into two parts: pre-branch and post-branch, and apply Lemma 4 on both parts, we have:

For $t \leq T$:

$$D_{TV}\left[p_i^t(s_i,a_i)\|\hat{p}_i^t(s_i,a_i)\right] \leq \varepsilon_{\pi_i}^{post} + \sum_{k=0}^{t-1} \varepsilon_{N_i^k}^{post},$$

and for $t > T$:

$$D_{TV}\left[p_i^t(s_i,a_i)\|\hat{p}_i^t(s_i,a_i)\right] \leq \varepsilon_{\pi_i}^{post} + \sum_{k=0}^{T-1} \varepsilon_{N_i^k}^{post} + \sum_{k=T}^{t-1} \varepsilon_{N_i^k}^{pre}.$$

Then, the difference of discounted distribution can be bounded as:

$$\begin{aligned}
D_{TV}\left[p_i(s_i,a_i)\|\hat{p}_i(s_i,a_i)\right] &\leq (1-\gamma)\sum_{t=0}^{\infty} D_{TV}\left[p_i^t(s_i,a_i)\|\hat{p}_i^t(s_i,a_i)\right] \\
&\leq (1-\gamma)\sum_{t=0}^{T}\gamma^t\left(\varepsilon_{\pi_i}^{post} + \sum_{k=0}^{t-1}\varepsilon_{N_i^k}^{post}\right) + \quad (1-\gamma)\sum_{t=T+1}^{\infty}\gamma^t\left(\varepsilon_{\pi_i}^{post} + \sum_{k=0}^{T-1}\varepsilon_{N_i^k}^{post} + \sum_{k=T}^{t-1}\varepsilon_{N_i^k}^{pre}\right) \\
&= \varepsilon_{\pi_i}^{post} + (1-\gamma)\sum_{k=0}^{T-1}\varepsilon_{N_i^k}^{post}(\gamma^{k+1}+\gamma^{k+2}+\ldots) + \quad (1-\gamma)\sum_{k=T}^{\infty}\varepsilon_{N_i^k}^{pre}(\gamma^{k+1}+\gamma^{k+2}+\ldots) \\
&= \varepsilon_{\pi_i}^{post} + \sum_{k=0}^{T-1}\varepsilon_{N_i^k}^{post}\gamma^{k+1} + \sum_{k=T}^{\infty}\varepsilon_{N_i^k}^{pre}\gamma^{k+1}.
\end{aligned}$$

We can convert this bound into the returns bound:

$$\begin{aligned}
|\eta_i - \hat{\eta}_i| &\leq \sum_{t=0}^{\infty}\gamma^t |r_i(s_i^t,a_i) - r_i(\hat{s}_i^t,\hat{a}_i^t)| \\
&\leq \frac{2r_{\max}}{1-\gamma}(1-\gamma)\sum_{t=0}^{\infty} D_{TV}\left[p_i^t(s_i,a_i)\|\hat{p}_i^t(s_i,a_i)\right] \\
&= \frac{2r_{\max}}{1-\gamma} D_{TV}\left[p_i(s_i,a_i)\|\hat{p}_i(s_i,a_i)\right] \\
&= \frac{2r_{\max}}{1-\gamma}\left[\varepsilon_{\pi_i}^{post} + \sum_{k=0}^{T-1}\varepsilon_{N_i^k}^{post}\gamma^{k+1} + \sum_{k=T}^{\infty}\varepsilon_{N_i^k}^{pre}\gamma^{k+1}\right]
\end{aligned}$$

Lemma 6. *(Returns bound measured in full length rollout) Consider two MDPs $p(s)$ and $\hat{p}(s)$. Suppose they both run their rollouts until the end of every trajectory, and the dynamic distributions are bounded as:*

$$\max_{s_{N_i},a_i} D_{TV}\left(p_i(s_i'|s_{N_i},a_i)\|\hat{p}_i(s_i'|s_{N_i},a_i)\right) = \varepsilon_{m_i}$$

while policy divergences are bounded as:

$$\max_{s_{N_i}} D_{TV}\left(\pi_i(a_i|s_{N_i})\|\hat{\pi}_i(a_i|s_{N_i})\right) = \varepsilon_{\pi_i}$$

Then the (local) returns are bounded as:

$$|\eta_i - \hat{\eta}_i| \leq \frac{2r_{\max}}{1-\gamma}\left[\varepsilon_{\pi_i} + \sum_{k=0}^{\infty}\gamma^{k+1}\sum_{j\in N_i^k}(\varepsilon_{m_j} + \varepsilon_{\pi_j})\right]$$

We can think it as a special case of branched rollout, where $\varepsilon^{post} = \varepsilon^{pre} = \varepsilon$ for every subscript, and $T = 0$. From this perspective, applying the result of Lemma 5 completes the proof.

5 Description of Supplementary Videos

5.1 Execution results in ATSC

We show the execution results of ATSC-Grid in our method (**Supplementary Video 1**) and DPPO (**Supplementary Video 2**). Through the key indicator "Insertion-backlogged vehicles" in the numerical panel on the left, we can observe that "Insertion-backlogged vehicles" will gradually increase with the load of the traffic flow, and then gradually decrease with the effective decision of the traffic light. The maximum value of this indicator is 1486 vehicles in DPPO and 1033 vehicles in our algorithm, which indicates that our approach can reduce the backlog of vehicles at intersections. Therefore, compared with DPPO, our framework is more effective in solving traffic jams. Through the three intersections we focused on in the video, we can also observe that our method can make traffic jams last for less time.

We further show a more challenging scenario with a heterogeneous network structure with diverse action and observation spaces: ATSC-Monaco traffic network with 28 intersections of real traffic network in Monaco city. We compare our method (**Supplementary Video 3**) with DPPO (**Supplementary Video 4**). Through the key indicator "arrived vehicles" in the numerical panel on the left, we can observe that " arrived vehicles " will gradually increase with the load of the traffic flow. The maximum value of this indicator is 563 vehicles in DPPO and 752 vehicles in our framework, which indicates that our method can reduce the backlog of vehicles at intersections and allow more vehicles to arrive at their destinations. Therefore, compared with DPPO, our algorithm is more effective in solving traffic jams. Through the three intersections, we focused on in the video, we can also observe that our method can make traffic jams last for less time.

5.2 Execution results in Flow

Supplementary Video 5 shows the execution result in Figure Eight. We can observe that our method controls the formation of queues to cross intersections with an orderly process of accelerating to the target velocity and then decelerating to the safe velocity near 0 m/s. We conclude that our models control vehicles to obey traffic rules while improving the efficiency of the overall traffic flow.

Supplementary Video 6 shows the execution result in Ring Attenuation. We can observe that our method can maintain the stability of the ring traffic flow and improve the efficiency of the overall traffic flow.

5.3 Execution results in Real Power-Net

Supplementary Video 7 shows the performance of the four algorithms on Real Power-Net with 141-bus, with particular emphasis on the total power loss. The observation period spans 12 hours, and for the sake of presentation, temporal augmentation has been applied to the video footage. The upper left, upper right, lower left, and lower right corners sequentially showcase the four algorithms: our method, CPPO, DPPO, and IC3Net. Through the video visualization, it becomes evident that our algorithm maintains the lowest aggregate power loss. The dynamic variation in colors of the power grid's structure signifies the dynamic changes in load, while the bar chart on the right provides a color scale corresponding to different loads; darker colors indicate higher load levels.

Supplementary Video 8 shows the performance of the four algorithms on Real Power-Net with 322-bus and more complex topology, with particular emphasis on the total power loss. The observation period spans 12 hours, and for the sake of presentation, temporal augmentation has been applied to the video footage. The upper left, upper right, lower left, and lower right corners sequentially showcase the four algorithms: our approach, CPPO, DPPO, and IC3Net. Through the video visualization, it becomes evident that our method maintains the lowest aggregate power loss. The dynamic variation in colors of the power grid's structure signifies the dynamic changes in load, while the bar chart on the right provides a color scale corresponding to different loads; darker colors indicate higher load levels.

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